

A distinctive meaning makes a sentence memorable

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Abstract

Prior work on visual memory has suggested that humans have a high-capacity but imperfect memory: image representations accumulate noise over time, which makes similar images confusable. This account — the *noisy representation hypothesis* — was recently extended to the verbal domain: in line with past evidence that words are encoded in memory by their meanings, it was shown that words with distinctive meanings are most memorable. Here, we leverage recent advances in natural language processing to ask whether the same holds true for compositional linguistic stimuli — sentences. In a recognition memory experiment with responses from 443 participants to 2,500 six-word-long target sentences, we found that a sentence’s semantic distinctiveness — as estimated with contextual representations from a large language model — predicts the accuracy and speed of its recognition. These effects were observed for both intrinsic sentence memorability (distinctiveness of a sentence relative to a large corpus of sentences) and contextual memorability (distinctiveness relative to recently encountered sentences in the experiment), and cannot be reduced to properties of the sentence’s constituent words. Our findings suggest that sentence memorability, similar to image and word memorability, is related to meaning distinctiveness, thus extending the noisy representation hypothesis to compositional linguistic stimuli.

Data availability

We share our data and analysis code at the following repository:
https://github.com/thomashikaru/sentence_memorability_share

1 Introduction

Every day, we come across objects, people, sounds and other experiences that feel familiar. In some cases, this familiarity has a basis in the underlying reality: an old acquaintance’s face looks familiar, or a song sounds familiar because we heard it on the radio a few times. In other cases, an item may seem familiar because we have come across something similar in the past, and

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our memories are not perfect. Previous work has identified numerous intrinsic properties of diverse stimuli — including visual objects, faces, sounds, and words — that make them more or less memorable (*intrinsic memorability*), and has also characterized contextual effects of recent experiences on recognition memory (e.g., Bainbridge, 2017; Bainbridge et al., 2013; Bartlett, 1977; Bower & Holyoak, 1973; Brady et al., 2008; Clément et al., 1999; Deutsch, 1970; Gorman, 1961; Kinsbourne & George, 1974; Konkle et al., 2010; Kramer et al., 2023; Lin et al., 2019; Revsine et al., 2024; Rust & Mehrpour, 2020; Standing, 1973; Steyvers & Malmberg, 2003; Tuckute et al., 2025; Wallace et al., 1970; Younesi & Mohsenzadeh, 2024). These empirical findings have led to various accounts of how we remember and forget, with hypotheses about both the nature of memory representations and the mental computations relevant to human recognition memory.

One candidate theory comes from the visual memory literature (see Brady et al., 2024, for a review). Under this account, items that are similar to one another are represented in similar portions of some representation space (the precise nature of this space may depend on the statistics of the relevant domain). Critically — given a finite memory capacity (Franconeri et al., 2013) — representations in memory accumulate noise over time, drifting from their original locations in the representation space. As a result, an item that falls in a high-density portion of the representation space (in terms of the concentration of other, similar items stored in memory) may be confusable with those of other, nearby items, leading to uncertainty about whether it has been encountered before. In contrast, an item that is distinctive (not similar to other items) may remain relatively easier to distinguish from other items in spite of the noise, leading to higher certainty about whether the item has been encountered before (Konkle et al., 2010; Markov et al., 2021; Rademaker et al., 2018; Wilken & Ma, 2004) (see illustration in Figure 1). This general idea aligns well with classic empirical findings and cognitive models that have highlighted the importance of distinctiveness for memorability (e.g., Brown et al., 2007; Nairne, 2006; von Restorff, 1933). Further, if memory representations are subject to noise, then probabilistic judgments of familiarity that take uncertainty into account may be the most efficient solution to making such judgments (Busemeyer & Townsend, 1993; Chater et al., 2006; Shiffrin & Steyvers, 1997); thus, the accuracy and speed of recognition may reflect graded levels of certainty rather than all-or-none memory, and forgetting may be attributed not to memories disappearing, but to interference from neighbors (Brown et al., 2007). We will refer to this account as the *noisy representation hypothesis*. A key advantage of this hypothesis is that it offers a unified explanation of phenomena in both **a**) long-term memory (having to do with how a given item relates to all other items in the relevant domain; denoted as *intrinsic memorability* effects), and **b**) short-term memory (having to do with how a given item relates to other items in the recent context, i.e. *contextual memorability* effects: Bays, 2014; Cowan, 2001; Gillies et al., 2023; Logie, 2014; van den Berg et al., 2012; cf. accounts that propose distinct mechanisms for long-term and short-term memory: Baddeley, 2003; Eichenbaum et al., 2007; Squire et al., 1993).

The goal of the current study is to evaluate the noisy representation hypothesis in the domain of memory for linguistic material. The bulk of past work on verbal memory has focused on single words, and one key finding that has emerged is that words are represented in memory primarily in terms of their meanings rather than their form-based features (Baddeley, 1966; Begg & Wickelgren, 1974; Franks & Bransford, 1972; Kroll & Potter, 1984; Potter & Lombardi, 1990; Tuckute et al., 2025). Building on this finding, Tuckute et al. (2025) provide support for the noisy representation hypothesis, showing that words with distinctive meanings — unambiguous words with few or no synonyms — are better remembered.

However, although words are the basic building blocks of language, much of human communication relies on more complex stimuli made up of multiple words — phrases and sentences. Some past studies have examined memory for sentences and established that, similar to words, sentences are encoded by their meanings: meaning-related information is better retained than the particular words that make up the sentence or the sentence’s syntactic structure (Arana et al., 2023; Begg & Wickelgren, 1974; Danescu-Niculescu-Mizil et al., 2012; Kroll & Potter, 1984; Potter & Lombardi, 1990). Yet, past work on sentence-level memory has been limited because, until recently, no tools existed for representing the meaning of arbitrary sentences, making it challenging to go beyond a small number of highly controlled items (e.g., Anderson, 1974) or simplistic representations of sentences (e.g., based on word or part-of-speech n-grams; Danescu-Niculescu-Mizil et al., 2012).

Recent advances in natural language processing and artificial intelligence have yielded large language models (LLMs) that can map entire sentences to a distributed, continuous semantic vector space (Reimers & Gurevych, 2019). LLMs, trained on massive language corpora with the objective of predicting an upcoming or a missing word, have been tremendously successful, achieving human-level performance on diverse language tasks (Brown et al., 2020; Wang et al., 2020). Armed with these new tools, we carried out a large-scale sentence memory experiment to test whether the distinctiveness of sentences within the meaning space — in general and with respect to recent experiences — explains how well they are remembered. A critical question we ask is whether compositional sentence-level meanings explain memory performance beyond the individual words that comprise the sentence. To foreshadow our key findings, we show that the semantic distinctiveness of a sentence relative to other sentences in the language is a strong predictor of recognition accuracy and decision speed, even after controlling for word-level memorability. Moreover, high semantic similarity of a sentence to recently encountered sentences is associated with slower responses and errors in the recognition judgments. Both of these effects support the noisy representation hypothesis, suggesting that this hypothesis can be successfully extended to the domain of compositional language.

2 Methods

2.1 Preregistration

The experimental design, sample size, and participant exclusion criteria were pre-registered on AsPredicted (ID: #104357), at the following URL: <https://aspredicted.org/4fsq-nnmb.pdf>.

2.2 Experiment overview

2.2.1 Design

We evaluated recognition memory for sentences in a behavioral experiment with 2,500 unique target sentences and $n=443$ participants.

Building on past work on image memorability (Bainbridge et al., 2013; Bylinskii et al., 2015; Goetschalckx et al., 2019; Isola et al., 2011a, 2011b; Khosla et al., 2016) and word memorability (Tuckute et al., 2025), the experiment was designed as a repeat detection task in which participants viewed a long sequence of sentences, presented one at a time, and were asked to press a

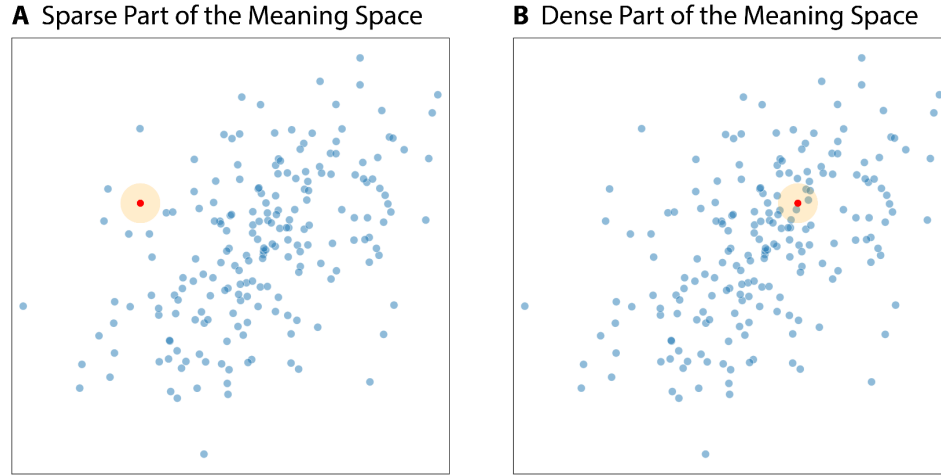


Figure 1: Illustration of the predictions of the noisy representation hypothesis for distinctive vs. less distinctive items. In this schematic, each dot represents an item (e.g., a sentence) encoded in a high-dimensional space (here, the meaning space; we show the space in two dimensions for ease of visualization). The red dot in each panel is the target sentence, for which a person must make a familiarity judgment. The orange disk represents uncertainty regarding the target sentence’s representation due to the accumulation of noise. We consider two cases: **(A)** The target sentence is located in a sparse part of the meaning space. Because this sentence has few near neighbors, when first encountered, it is less likely to be mistaken for a previously encountered sentence (a low probability of a false alarm). On a repeat encounter, there is less likely to be interference from previously encountered sentences, because there are no other sentences that could have generated this memory representation (a high probability of a hit). **(B)** The target sentence is located in a dense part of the meaning space. Because this sentence has many near neighbors, when first encountered, it is more likely to be mistaken for a previously encountered sentence (a high probability of a false alarm). On a repeat encounter, there is more likely to be interference from previously encountered similar sentences (a low probability of a hit). See Section 4.3 for a discussion about whether the space is constructed through the context of the experiment and/or general language experience.

key whenever they noticed a repeat (i.e., a sentence that had appeared earlier in the sequence). Each participant saw a sequence of 1,020 sentences, divided into target sentences (each of which appeared twice, separated by ~100 sentences, as detailed below) and filler sentences (which only occurred once or served as vigilance repeats, separated by 1-7 trials, as detailed below).

2.2.2 Participants

Participants were recruited using the Prolific crowd-sourcing platform. The study was restricted to participants based in the United States who listed English as their native language. Participants were also asked about their native language, and were told that payment would not be contingent on their answers; data from participants who did not self-report English as their native language was excluded. 500 participants took part in the experiment, with 443 remaining after pre-defined exclusion criteria (Section 2.4.1). The experiments were conducted with approval from and in accordance with the Committee on the Use of Humans as Experimental Subjects at the Massachusetts Institute of Technology. Participants provided informed consent before starting the experiment and were compensated for their time (US\$12, total experiment duration was approximately one hour).

2.3 Materials

For the target sentences, a total of 2,500 six-word-long sentences were extracted from naturalistic corpora (see Section 2.3.1). Note that although not every stimulus constitutes a sentence by some definitions (some stimuli are fragments or headlines), we use the term ‘sentence’ throughout for simplicity. 1,500 of these target sentences were selected so as to fall into three conditions ($n=500$ sentences each) based on the memorability of their constituent words — LOW, MEDIUM, and HIGH word-level memorability — as detailed below. The remaining 1,000 target sentences in the DIVERSE condition were chosen without any constraints on word-level properties and were included to sample a broad, naturalistic set of sentences irrespective of word-level memorability.

2.3.1 Source corpora

We aimed to include semantically, syntactically, and stylistically diverse sentences. To do so, we selected six-word sentences from the following three main categories of corpora:

i) Written-published:

- The Brown corpus contains written American English text from 14 different genres, such as newswire, memoirs, and fiction. The Brown Corpus was created in 1961 (accessed via NLTK: Bird et al., 2009).
- The Wall Street Journal corpus contains American newswire from the Wall Street Journal (Paul and Baker, 1992; available as a subset of <https://catalog.ldc.upenn.edu/LDC95T21>).
- The Toronto BookCorpus (Zhu et al., 2015) is a large collection of amateur fiction from which we used three different genres: Fantasy, Adventure, and Mystery.

ii) Written-web:

- The Common Crawl C4 (Raffel et al., 2023) is a massive collection of English text from the internet, containing data up to 2019.

iii) Spoken:

- The Cornell Movie Dialogues (Danescu-Niculescu-Mizil et al., 2012) is a large collection of conversations extracted from raw movie scripts dated up to 2011.
- The Corpus of Contemporary American English (COCA; Davies, 2010) contains sentences from a variety of genres, but here, we used the spoken subset. The spoken subset contains transcripts of unscripted conversation from TV and radio programs, covering sources from 1990 to 2012.

For the DIVERSE set, a few additional smaller corpora were included in the selection process, as detailed in Tuckute et al. (2024). Sentences were preprocessed to remove non-standard punctuation, capitalization, and other artifacts. For details, see Section SI 1.

2.3.2 Selection of the sentences for the LOW, MEDIUM, and HIGH memorability conditions

Sentences were selected for the LOW, MEDIUM, and HIGH conditions based on the memorability of their constituent words. For a subset of words, empirical memorability was known from prior work on word memorability (Tuckute et al., 2025). The selection proceeded in four steps. First, from among words with empirically established memorability ($n=2,783$ single words), three subsets of high, medium, and low empirical memorability were sampled, resulting in 1,428 “ground-truth” words. Next, a linear regression model based on GloVe features (Pennington et al., 2014) was trained to estimate word memorability for arbitrary words within the GloVe vocabulary, using the available empirical word memorability data as a training set. Finally, experimental stimuli were selected based on the presence of words of known memorability and overall estimated memorability — for example, sentences for the HIGH memorability condition contained at least one of the empirically highly memorable words, and also had a high average estimated word memorability (see Section SI 2 for details). Example sentences with highly memorable words included “*Goddammit, where are you from, Mars?*” and “*These mosquitos are -- well, guinea pigs.*”. Example sentences with no memorable words included “*Actually, I already work for you.*” and “*He has a very powerful ship.*”. These steps yielded three sets of sentences that were categorized according to estimated word memorability, without considering how words in the sentences combined compositionally.

2.3.3 Selection of the sentences for the DIVERSE condition

The DIVERSE condition consisted of 1,000 six-word sentences. Approximately half of the sentences were selected to be semantically diverse, spanning 180 semantic clusters as identified in Pereira et al. (2018) using the GloVe word embedding space (Pennington et al., 2014); the other half were pseudo-randomly sampled from diverse text corpora to ensure stylistic diversity. These 1,000 sentences were selected without taking any memorability criteria into account. The detailed selection procedure is described in Tuckute et al. (2024).

2.3.4 Selection of filler sentences

1,000 six-word filler sentences were sampled from the same source corpora as the target sentences, but with the constraints of not containing any of the 1,428 words in the empirical word memorability dataset used to select sentences for the LOW, MEDIUM, and HIGH conditions. For details, see Section SI 2.

2.3.5 Word edit distance filtering and manual review

To ensure that no sentences in the full set of 3,500 (2,500 targets and 1,000 fillers) were identical or close to identical, we filtered out sentences based on word edit distance: given two sentences, the word edit distance between them is the minimum number of insertions, deletions, or substitutions of whole words required to transform one sentence into the other, ignoring punctuation and capitalization. We identified all pairs of sentences with a word edit distance less than 3, and kept only one sentence from each pair (with the constraint of preserving the set of 1,000 sentences in the DIVERSE condition as this set of materials was used in an ongoing, parallel study). Finally, the remaining sentences (1,768 critical sentences and 1,175 filler sentences) were manually reviewed for typographical errors and potentially offensive content (see Section SI 1 for the exclusion criteria), and a final set of 1,500 critical sentences (500 sentences for the LOW, MEDIUM, and HIGH word memorability conditions) and a set of 1,000 filler sentences were selected for use, in addition to the 1,000 sentences from the DIVERSE condition.

2.3.6 Creation of experimental lists

The experimental materials in each condition (500 sentences in each of LOW, MEDIUM, and HIGH memorability conditions, and 1,000 sentences in the DIVERSE condition) were assigned to 12 random and disjoint subsets: HIGH_{1-12}, MEDIUM_{1-12}, LOW_{1-12}, and DIVERSE_{1-12}. Each subset in the LOW, MEDIUM, and HIGH conditions contained 41 or 42 sentences (500 / 12), whereas each subset in the DIVERSE condition contained 83 or 84 sentences (1,000 / 12). For each participant, the set of critical sentences was assembled by randomly selecting one subset from each of the four conditions (HIGH, MEDIUM, LOW, DIVERSE), i.e., a total of 206-210 critical sentences; the remaining sentences in the list were sampled from the set of 1,000 filler sentences, for a total of 1,020 trials. In this way, approximately 40% of the 1,020 trials in a given list were critical sentences, divided between the DIVERSE (~16%), LOW (~8%), MEDIUM (~8%), and HIGH (~8%) conditions, and the remaining ~60% were fillers.

2.4 Procedure

The sentences appeared one at a time, and participants were asked to press the SPACE key when a sentence occurred that they had already seen. Each sentence was presented for 2 seconds followed by a 1.4 second fixation, giving participants 3.4 seconds to respond (Figure 2A). The order of the 1,020 sentences in a given list was determined by a sequence generation procedure (adapted from Tuckute et al., 2025) which outputs a pseudo-randomized sequence with the following properties (Figure 2A):

i) Repeats of the target sentences (each target sentence is presented exactly twice) occur at an interval of 91-109 sentences. Consequently, ~20% of trials are target repeats, i.e., the second

presentation of a critical sentence. In the dataset we collected, each target sentence was seen twice by at least 19 and at most 49 participants (median=35; mean=35.4). The median number of DIVERSE trials across participants was 154 (mean=149), and the median number of trials in each of the LOW, MEDIUM, and HIGH conditions was 77 (mean=74).

ii) 75% of the filler sentences (45% of the total trials in the list) occur only once, and the remaining 25% of the filler sentences (15% of the total trials in the list) are ‘vigilance’ repeats, which occur at short intervals (1-7 trials) after the initial occurrence. These vigilance repeats served as an attention check.

The experiment took approximately one hour. Participants were expected to complete the entire experiment, but could take a short break after every block of 85 trials (each list was divided into 12 such blocks). The maximum time allowed for a break was 3 minutes, indicated by a countdown timer, at the end of which the experiment continued automatically; however, the median break length taken by participants was just 10.6 seconds. Throughout the experiment, a progress bar at the top of the screen indicated the participant’s progress in the experiment and was updated after every trial.

2.4.1 Participant exclusion criteria

The following exclusion criteria were pre-registered before collecting the experimental data. Participants were excluded from the analyses if: **a)** the vigilance hit rate for the entire experiment was lower than 0.5, or **b)** the false alarm rate for the entire experiment was higher than 0.25. And an individual block of trials for a participant was excluded from the analyses if: **a)** the vigilance hit rate for that block was lower than 0.5, or **b)** the false alarm rate for that block was higher than 0.25.

2.5 Analysis and modeling

2.5.1 Memorability measures

For each critical sentence, we define three empirical measures of memorability: hit rate (proportion of participants who correctly reported the sentence as a repeat when encountering it the second time), false alarm rate (proportion of participants who incorrectly reported the sentence as a repeat when encountering it for the first time), and overall response accuracy (proportion of all trials, across participants, which were either correct detections or correct rejections). The accuracy metric reflects how likely a given sentence is to be correctly recognized while also penalizing false alarms, and is intuitively easily interpretable. Previous studies have sometimes used d' as a measure of memorability, although typically for quantifying an individual participant’s performance, not as an item-level property. Because d' relies on taking the z-transforms of the hit and false alarm rates, it is undefined when these values are either 0 or 1. When excluding these cases, d' has a Pearson correlation of 0.945 ($p < 0.001$) with our accuracy measure. When using a version of d' that omits the z-transform, d' has a Pearson correlation of 0.997 ($p < 0.001$) with the accuracy measure. Henceforth, we use “memorability” to refer to response accuracy, consistent with the terminology of Tuckute et al. (2025) at the word level.

2.5.2 Predictors of intrinsic memorability

To evaluate the predictions of the noisy representation hypothesis, for each sentence, we generated predictors based on the memorability of constituent words and critically, based on the distinctiveness of sentence-level meanings. When examining the word-level memorability effects, we controlled for potential influences of word frequency — a feature that has been implicated in several past studies investigating memory for words (e.g., Gorman, 1961; Hulme et al., 1997; Khanna & Cortese, 2021; Rubin & Friendly, 1986).

Word-level memorability We estimate a memorability score for each sentence based on the memorability of its constituent words, obtained as described in Section 2.3.2 and Section SI 2. There are many *a priori* plausible ways in which the memorability of individual words in a sentence could affect its overall memorability. The simplest version of this hypothesis is that the *average* memorability of the words predicts sentence memorability, but other plausible variants include the *maximum* memorability, or the memorability of words in particular sentence positions (e.g., at the beginning or end of the sentence, given the classic serial position curve effects; Ebbinghaus, 1913; Murdock Jr., 1962). To select the best predictor of estimated word memorability, we conducted a predictor performance analysis, cross-validated over held-out participants and sentences, on the DIVERSE set consisting of 1,000 linguistically diverse, naturalistic sentences (Section SI 3). We found that average word memorability best predicted sentence memorability, and hence we use this predictor in the main analyses. The theoretically possible range of the average word memorability score goes from 0 to 1; in practice, the lowest estimated score for a target sentence was 0.71 (“*Not just physically but intellectually, emotionally.*”), and the highest was 0.85 (“*Herbed chevre, pomodoro sauce, basil toast.*”). Importantly, for this metric, any word in the sentence could be replaced with a word that has an equivalent memorability score without affecting the sentence memorability score, so this metric is not sensitive to properties that emerge from composition or co-occurrence of words in the sentence.

Word frequency and length covariates We compute the average log frequency of the words in a sentence based on the SUBTLEXus database (Brysbaert & New, 2009). This corpus provides frequencies (per million words), calculated based on a 51-million word corpus of movie subtitles. The genre of movie subtitles was chosen to more closely resemble everyday language use, as opposed to prior word frequency norms that used less naturalistic corpora and smaller sample sizes (Baayen et al., 1993; Kucera & Francis, 1967); this choice is supported by a better fit to human lexical decision times using the SUBTLEXus norms compared to previous norms. Words that do not occur in the SUBTLEXus corpus are automatically assigned the minimum frequency value from SUBTLEXus of 0.02 occurrences per million words. For words which SUBTLEXus breaks into multiple tokens, such as contractions (e.g., “can’t”), we average the frequencies of subword tokens. Punctuation and capitalization were ignored for the purpose of computing word frequencies (the SUBTLEXus norms are case-insensitive). After computing the average log word frequency of each sentence, the 2,500 critical sentences in our experiment had a mean value of 5.84, a median of 6.09 (the sentences “*He handed her the leather bag.*” and “*Sounds just a little complicated, appropriate?*”), a minimum of -0.86 (the sentence “*Herbed chevre, pomodoro sauce, basil toast.*”), and a maximum of 9.29 (the sentence “*I told you what to do.*”). To translate these values into more interpretable units, the lowest-scoring sentence by this metric uses words that

appear, on average, once every 2.4 million words, while the highest-scoring sentence uses words that appear, on average, once every 92 words. For reaction time analyses, in order to factor out variance due to word lengths, we additionally include a covariate of the total sentence length in characters.

Sentence-level semantic distinctiveness As discussed in Section 1, recent advances in natural language processing allow mapping words and sentences to semantic vector spaces. Such spaces allow us to measure how distinctive a sentence is relative to (some set of) other sentences. Specifically, we compute a vector representation for each sentence and calculate the mean cosine distance between this sentence’s vector and a large random sample of sentences drawn from the same corpora as the experimental stimuli (i.e., the full set of 60,000 sentences described in Step 2 in Section 2.3.2). This distinctiveness metric is a scalar value in the interval of 0 to 2, with higher values indicating that a given sentence’s representation is more distinctive (i.e., less similar to those of other sentences in the set). This metric provides an interpretable notion of how unique a sentence is in an abstract high-dimensional semantic space. We compute the distinctiveness measure based on sentence embeddings from six word-level or sentence-level language models, which differ in their training objectives. In the analyses, we compare these distinctiveness scores to identify the best predictor of empirical sentence memorability.

GloVe Sentence Embedding Distinctiveness: We obtained a 300-dimensional GloVe vector (Pennington et al., 2014) for each of the words in each sentence, and averaged the vectors across the words in the sentence to compute a sentence-level representation. This simple aggregation of word embeddings treats sentences as bags of words: any two sentences containing the same words in different orders would receive the same distinctiveness score, even if their meanings are different. The following sentences had the 5 highest GloVe distinctiveness scores across all experimental materials: “*Tantalum, Titanium, Hastelloy, Inconel and Monel.*”, “*Herbed chevre, pomodoro sauce, basil toast.*”, “*Bandits, rustlers, Texas rednecks, Kiowa, Comanche.*”, “*That’s cuckoo; it’s nuts; it’s broke.*”, “*Tubs, cartons, soap pumps, shampoo bottles.*” Many of these examples are simply word lists consisting of diverse and unusual words.

SBERT Sentence Embedding Distinctiveness: We obtained contextualized sentence representations from the SBERT sentence embedding model bert-base-nli-mean-tokens (Reimers & Gurevych, 2019). SBERT, which stands for Sentence-BERT is trained to map sentences to representations that are useful for various downstream language tasks. We used the pretrained model, with the default parameters, available via the Sentence Transformers library (sentence-transformers version 2.2.2). The following sentences had the 5 highest SBERT distinctiveness scores across all experimental materials: “*Texas steak dinner on a stick?*”, “*Alaska banned wolf hunting by airplanes.*”, “*Vegetable soup and a turkey sandwich.*”, “*My son was outside playing soccer.*”, “*I’m making enchilada pasta for dinner.*” Qualitatively, these sentences express concrete meanings, with or without low-frequency or unusual words.

GPT-2 Sentence Embedding Distinctiveness: We obtained contextualized sentence representations from the unidirectional-attention Transformer model gpt2-medium (Radford et al., 2019). This model has 24 layers and an embedding dimension of 768. We tokenized each sentence using the model’s standard tokenizer (GPT2TokenizerFast) and retrieved model activations for each model layer. We obtained the representation of the last sentence token, given that unidirectional models already aggregate representations of the preceding context (i.e., earlier tokens in

the sentence). We used the pretrained model available via the HuggingFace library (Wolf et al., 2020), and we report results for the best-performing model layer, which was layer 23 (see Section SI 7 for results based on other layers of GPT-2). The following sentences had the 5 highest GPT-2 distinctiveness scores across all experimental materials: “*I think it’s overblown,’ he said.*”, “*Robert Boochever, Juneau, Alaska, for appellees.*”, “*Q: What is 147451 times Pi?*”, “*Free puck retrieved by Matt Niskanen.*”, “*For NPR News, I’m Nancy Solomon.*”. Qualitatively, these examples tend to contain many proper nouns.

BERT Sentence Embedding Distinctiveness: We obtained contextualized sentence representations from the bidirectional-attention Transformer model bert-large-cased (Devlin et al., 2019). This model has 24 layers (i.e., Transformer blocks) and an embedding dimension of 768. We tokenized each sentence using the model’s standard tokenizer (BertTokenizerFast) and retrieved activations for each model layer. We obtained the representation of the special [CLS] token which is prepended to the sequence and is standardly used as a token for classification output (Devlin et al., 2019). We used the pretrained model available via the HuggingFace library (Wolf et al., 2020), and we report results for the best-performing model layer, which was layer 9 (see Section SI 7 for results based on other layers of BERT). The following sentences had the 5 highest BERT distinctiveness scores across all experimental materials: “*A scream spilled from Kylie’s lips.*”, “*Burnett continued staring right at her.*”, “*I’ll send Mendigo to fetch Dale.*”, “*Karina swept Emily into her arms.*”, “*Niol licked his lips, tasted her.*”. Qualitatively, these examples tend to contain vivid emotional language.

ELECTRA Sentence Embedding Distinctiveness: We obtained contextualized sentence representations from the ELECTRA sentence embedding model (Clark et al., 2020). The following sentences had the 5 highest ELECTRA distinctiveness scores across all experimental materials: “*One, two, three, four, five, six-*”, “*NOTD: Collection 2000 hot looks ‘Diva.’*”, “*Sizes range from 2XL to 3XL.*”, “*Estimated electricity use in residential sector.*”, “*Passport needed for international boat cruise?*”.

Universal Sentence Encoder (USE) Sentence Embedding Distinctiveness: We obtained contextualized sentence representations from the Universal Sentence Encoder model (Cer et al., 2018). The following sentences had the 5 highest USE distinctiveness scores across all experimental materials: “*Implement continuous and routine improvement ideas.*”, “*Smart color control and modular structure.*”, “*Gator category: microphone, wireless system cases.*”, “*Ideal for indoor or outdoor cultivation.*”, “*For more details, see Sharpen photos.*”.

These six distinctiveness measures had a mean pairwise correlation (computed across the 1,000 DIVERSE sentences) of 0.26 (SD = 0.12), which suggests that they may capture different aspects of sentence meanings. ELECTRA and SBERT distinctiveness scores showed the lowest correlation (0.07), whereas GloVe and USE scores showed the highest (0.49). Correlations between all pairs of sentence memorability predictors are shown in Section SI 10.

2.5.3 Accuracy analysis

We tested whether semantic distinctiveness is a significant predictor of accuracy. As elaborated just above, we operationalize semantic distinctiveness according to a range of models; to assess the increase in predictive power attributable to a given semantic distinctiveness predictor, we perform a delta log-likelihood (DLL) analysis comparing the change in the likelihood of the observed data when adding a target predictor to a baseline linear regression model which only

includes control predictors. The change in log-likelihood when including the target predictor is divided by the number of observations to yield a per-observation Delta log-likelihood score. We repeat this procedure across 10 folds of data to arrive at a mean and 95% confidence interval for the Delta log-likelihood score. This type of analysis has been utilized to compare the predictive power of a set of candidate predictors of a given psychometric variable, such as reading times or regressive eye movements in reading (Goodkind & Bicknell, 2018; Wilcox et al., 2024). This approach is especially relevant when comparing a number of related predictors, such as semantic distinctiveness computed using different methods. Regressions were fit using the `lm` function in R (R Core Team, 2022), and all continuous variables were centered and scaled.

2.5.4 Reaction time analysis

On trials where participants endorsed a critical sentence as a repeat (correctly or incorrectly), we analyzed the reaction times measured from the beginning of the presentation of the sentence to when the space bar was pressed. Critically, we tested whether semantic distinctiveness is a significant predictor of reaction time. Because reaction times depend, in part, on how long a stimulus takes to read, we controlled for variables known to affect reading times (Brothers & Kuperberg, 2021; Demberg & Keller, 2008; Hahn & Keller, 2022; Kennedy et al., 2013; Luke & Christianson, 2018; Smith & Levy, 2013; Wilcox et al., 2023): average word frequency and sentence length in characters (all sentences had the same number of words); we also controlled for word memorability and repeat status (whether the item had appeared previously). We fit a linear mixed-effects regression model using the `lmer()` function from the *lme4* package in R (Bates et al., 2015), with random intercepts for participants and sentences. All continuous variables were centered and scaled.

2.5.5 Predictors of contextual memorability

To understand how features of previously encountered stimuli may affect participants' recognition memory for a particular sentence (encountered for the first time), such that they incorrectly believe that they had seen this sentence before, we computed two additional predictors based on the relationships among the sentences in the unique sequence used for each participant. First, to investigate the effect of word-level properties, we quantified the degree of lexical overlap in the materials. For each participant, for each trial where a false alarm is possible, we computed a measure of content word overlap — a binary variable denoting whether any content word in the current sentence had appeared in any previous sentence in the experiment. We restricted this measure to content words, defined as words not present in the NLTK library's English stopwords collection (Bird et al., 2009), because most function words, such as “*the*” or “*and*”, occur in many stimuli. Content word overlap is likely to be a significant predictor of false alarms if participants rely on the presence of a repeated word as evidence of a repeated sentence, even if the word occurs in a novel sentence.

Second, to investigate the effect of sentence-level properties, we quantified the similarity of sentence-level meanings in the materials. For each participant, for each trial where a false alarm is possible, we computed the cosine similarity to each sentence in the preceding sequence, using representations from the SBERT sentence embedding model (given that, as will be discussed below, the semantic distinctiveness measure based on this model yielded the best performance in

the accuracy analysis), and identified the sentence with the highest value. We refer to this predictor as the distractor cosine similarity, as it represents a sentence’s similarity to the most similar previously encountered sentence. For example, the two sentences in our set “*We all enjoyed it very much.*” and “*Everyone was so impressed by it.*” had a distractor cosine similarity score of 0.95. Meanwhile, a more dissimilar pair of sentences — “*So yes, we are considering alternatives.*” and “*That’s what I been thinkin’ about.*” — had a distractor cosine similarity score of 0.78. The presence of a significant effect of distractor cosine similarity on false alarms after controlling for content word overlap would suggest that sentences with similar meanings, even when they don’t share content words, are confusable in memory, which would imply that sentences are encoded in memory by their compositional meanings and not just using the representations of their constituent words.

We fit a logistic mixed-effects regression model to predict responses on target repeat trials (i.e., trials where false alarms are possible) using the two critical confusability-based predictors (content word overlap and distractor cosine similarity), controlling for the sentence-intrinsic features of average word memorability and average word frequency, using the `glmer()` function from the *lme4* package in R (Bates et al., 2015). All continuous variables were centered and scaled. Random intercepts were included for participants and items.

2.6 Intuitive judgments of sentence memorability

Alongside collecting empirical sentence memory data, we also asked whether individuals have an intuitive sense of which sentences are more memorable. For the DIVERSE sentence set ($n=1,000$), we collected subjective judgments of memorability from an independent set of participants ($n=400$, with $n=293$ after exclusions, with each sentence rated by 12-17 participants). Participants were told that a memorable sentence is likely to be remembered a few hours later, but a non-memorable sentence is likely to be quickly forgotten. They rated each sentence on a Likert scale from 1 (not at all memorable) to 7 (highly memorable) (see Section SI 9 for details).

3 Results

3.1 Memorability is a consistent property of sentences

In a large-scale behavioral recognition memory experiment (see overview in Figure 2A) with 2,500 target sentences and 443 participants, the median accuracy (see Section 2.5.1) for target sentences was 80.2%, the hit rate (correct detections) was 64.3%, and the false alarm rate (incorrect detections) was 4.3% (Figure 2B). Importantly, there was substantial variation in memorability across sentences, with accuracies ranging from 46.0% (the most forgettable sentence) to 100% (the two most memorable sentences; see Figure 2C for sentence examples), hit rate ranging from 3.4% to 100%, and false alarm rate ranging from 0% to 35.6%.

We evaluated the consistency of memory accuracy (henceforth ‘sentence memorability scores’) across participants by dividing our participant pool in half and computing a correlation across sentences to ask: if a sentence is memorable for one half of participants, is it also memorable for the other half of participants? The median Spearman’s ρ across 1,000 random half-half participant splits was 0.56 (95% CI: 0.52-0.60) for accuracy, 0.56 (95% CI: 0.52-0.60)

for hit rate, and 0.24 (95% CI: 0.21-0.28) for false alarm rate. This consistency indicates that some sentences are remembered better than others in a consistent way across participants who saw sentences in randomized orders, suggesting that some intrinsic sentence properties affect memorability. The relatively low split-half correlation in false alarm rate suggests that false alarms are affected more strongly by context (i.e., the order of items, which varied across participants) rather than intrinsic sentence properties (see Section 3.4 for a detailed analysis of false alarms and their relevance for the noisy representation hypothesis).

3.2 Memorability of individual words predicts intrinsic sentence memorability

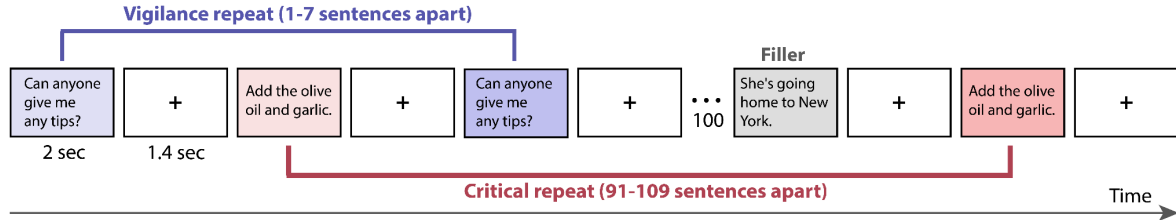
Having established that accuracies on the sentence memory task are high and that some sentences are consistently more memorable than others, we investigate the nature of sentence memory representations by first asking how well the memorability of the individual words that comprise the sentence predicts sentence-level memorability (see Section 2.5.2).

Qualitatively, the most versus least memorable sentences (Figure 2D, Section SI 5) appear to differ in the types of words that they contain: more memorable sentences tend to contain more content words, low-frequency words, and proper nouns. To quantitatively evaluate these differences, we performed two analyses. First, we compared the three experimental conditions that vary in average word-level memorability (see Section 2.5.2). The HIGH word-level memorability condition comprised sentences that contain memorable words, whereas the LOW word-level memorability condition consisted of sentences which only contain not very memorable words. The data suggest that sentences in the HIGH condition are more memorable, on average, than those in the MEDIUM condition (Mann-Whitney U-test: 71871.0, $p < 0.001$), and those, in turn, are more memorable than the sentences in the LOW condition (Mann-Whitney U-test: 67415.5, $p < 0.001$). Figure 3A shows mean accuracy by condition, while Figure 3B shows the relationship between mean word memorability and sentence memorability across target sentences in the DIVERSE condition (see Section SI 4 for hit rate and false alarm rate by condition; see Section SI 5 for the most and least memorable sentences by condition).

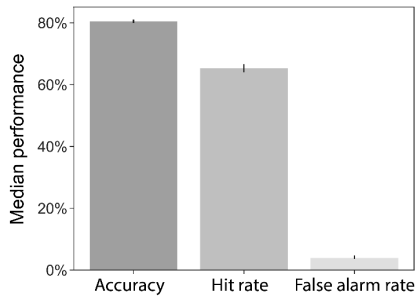
Second, we evaluated how well the predictor of estimated word-level memorability explains sentence memorability in the DIVERSE set of 1,000 sentences — a set of sentences that were selected without any consideration of memorability. The data suggest that the estimated word-level memorability of a sentence is a significant predictor of sentence memorability ($\beta = 0.476$, $SE = 0.04$, $p < 0.001$; Figure 3B), even when controlling for word frequency — a feature known to influence both word- and string-level memorability (Brown & Lewis, 1981; Gorman, 1961; Kinsbourne & George, 1974; Lohnas & Kahana, 2013; Popov et al., 2022; Schulman, 1976). Hence the memorability of the constituent words robustly affects sentence-level memorability.

Finally, we show that word-level memorability also affects reaction times. In a linear mixed-effects model fit on all trials for which participants reported a sentence as familiar, word memorability was found to lead to faster reaction times ($\beta = -46$, $SE = 3.1$, $p < 0.001$), even when controlling for word frequency, sentence length (known to contribute to reading times for sentences; e.g., Demberg and Keller, 2008), and whether the response constituted a correct detection or a false alarm.

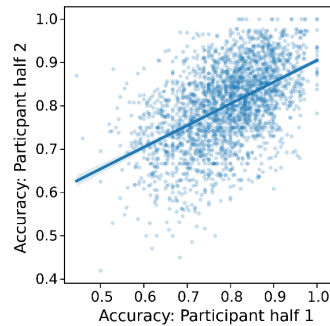
A Schematic of Experimental Paradigm



B Memorability of Sentences



C Consistency of Sentence Memorability Scores Across Participants



D Most and Least Memorable Sentences

Most memorable

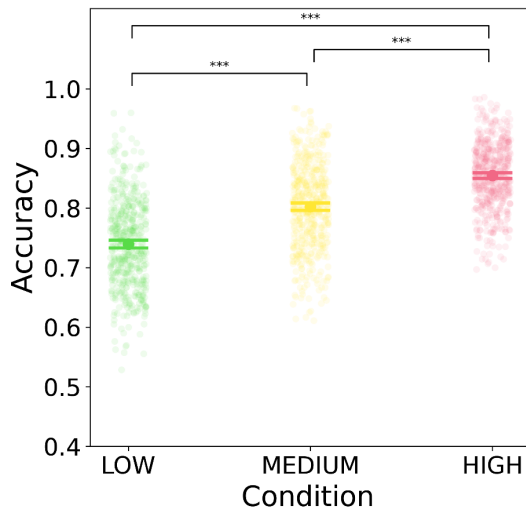
Wait a sec, Nicky, Nicky, Nicky.
Homer Simpson is hungry, very hungry.
Does olive oil work for tanning?
These mosquitos are -- well, guinea pigs.
Goddammit, where are you from, Mars?
Macbook Pro repair in Niagara region.
They stabbed her fifty three times!
Now let's talk about Donkey Kong.
Top with mozzarella, mushrooms, and peppers.
Every cloud has a blue lining!

Least memorable

I just couldn't leave without it.
We want to make it better.
But still it wasn't nearly enough.
It must be getting more serious
But it made me feel better..
Therefore to hold meetings by them.
Another manner is to time entries.
But give it to me again.
You still had to prove yourself.
Unless he'd been afraid they'd know.

Figure 2: Experimental paradigm, memory performance, and most and least memorable sentences. **(A)** Illustration of the experimental paradigm. Each sentence was presented for 2s, with a 1.4s inter-trial interval, providing participants 3.4s to respond. Participants were asked to press the SPACE key when they encounter a sentence that had already occurred at some point in the experiment. Critical memory repeats occurred between 91 and 109 sentences apart, and vigilance repeats occurred between 1 and 7 sentences apart. **(B)** Median response accuracy, hit rate, and false alarm rate, averaged across the 2,500 target sentences ($n=443$ participants). Error bars indicate bootstrapped 95% confidence intervals. **(C)** Consistency of the sentence memorability accuracy across participants. The scatter plot shows the mean accuracy across all sentences ($n=2,500$) for half of the participants versus the mean accuracy from the other half for one split of the data. **(D)** Most memorable and least memorable 10 sentences from the HIGH, MEDIUM, and LOW conditions, based on accuracy.

A Sentences Grouped by Word Memorability



B Diverse Sentences Selected Without Taking Word Memorability Into Account

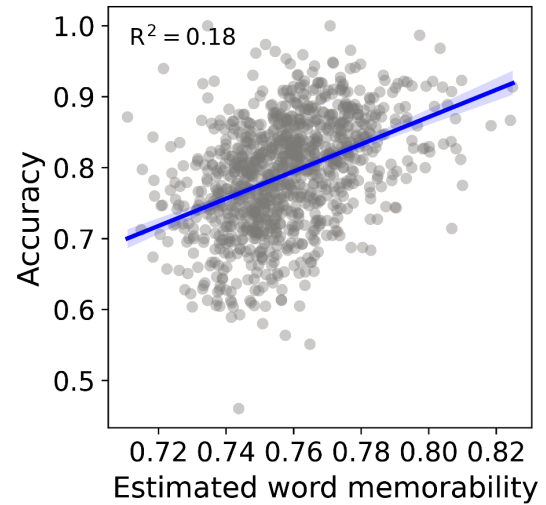


Figure 3: The relationship between word memorability and sentence memorability. **(A)** Average response accuracy of sentences in the LOW, MEDIUM, and HIGH word memorability conditions. Error bars indicate 95% confidence intervals. Each dot represents one sentence. The presence of memorable words makes sentences more memorable. **(B)** Average response accuracy of sentences in the DIVERSE condition plotted as a function of average estimated word memorability, showing a positive correlation. Each dot represents one sentence. Line of best fit is shown, with shaded areas indicating the 95% confidence interval.

3.3 Sentence meaning distinctiveness predicts intrinsic sentence memorability, especially for sentences with less memorable words

In the previous section, we demonstrated that word-level memorability is a significant predictor of sentence memorability. Does the distinctiveness of sentence-level meanings explain sentence memorability beyond the effect of word-level memorability? And how do word-level memorability and sentence meaning distinctiveness interact? As described in Section 2.5.2, we compare several semantic distinctiveness metrics, each of which assigns a scalar value to a sentence based on how unique the sentence is in an abstract high-dimensional vector space obtained via different language models. We focus here on the sentences in the DIVERSE condition in order to capture a wide range of sentence types which vary continuously in their word- and sentence-level properties, unlike the discretized sentences in the HIGH, MEDIUM, and LOW word memorability conditions. We compare the predictive power of the semantic distinctiveness measures to each other, while controlling for word-level memorability and word frequency, following the Delta log-likelihood procedure (Section 2.5.3). Figure 4 shows the Delta log-likelihood (DLL) scores across the six measures. Although all distinctiveness measures show positive DLL, distinctiveness derived from the language model SBERT fares best (see Section SI 6 for the full statistical results), with distinctiveness computed using other models such as USE and ELECTRA having much lower predictive power for memorability.

Critically, we observe a significant effect of SBERT distinctiveness ($\beta = 0.283$, $SE = 0.030$, $p < 0.001$) on accuracy, even when controlling for word-level memorability ($\beta = 0.344$, $SE = 0.040$, $p < 0.001$) as well as word frequency ($\beta = -0.126$, $SE = 0.031$, $p < 0.001$). This finding demonstrates that semantic distinctiveness (in this case, quantified through the SBERT representational space) is an important predictor of sentence memorability, explaining variance that is not accounted for by word-level features alone. We also observe a significant negative interaction between semantic distinctiveness and word-level memorability ($\beta = -0.161$, $SE = 0.034$, $p < 0.001$), such that semantic distinctiveness has a larger effect for sentences that do not contain memorable words.

Finally, we show that semantic distinctiveness also affects reaction times. In a linear mixed-effects model fit on all trials for which participants reported a sentence as familiar, semantic distinctiveness was found to lead to faster reaction times ($\beta = -36$, $SE = 2.9$, $p < 0.001$), even when controlling for word-level memorability ($\beta = -46$, $SE = 3.1$, $p < 0.001$), word frequency ($\beta = -22$, $SE = 3.9$, $p < 0.001$), sentence length ($\beta = 9.8$, $SE = 0.6$, $p < 0.001$), and whether the response constituted a correct detection or a false alarm ($\beta = -199$, $SE = 5.7$, $p < 0.001$) (see Section SI 6 for the full statistical results).

In summary, in line with the noisy representation hypothesis, semantic distinctiveness shows clear effects even when controlling for word-level memorability on both accuracies and reaction times.

3.4 Word overlap and sentence-meaning similarity with recently encountered sentences explain contextual memorability effects

In the previous sections, we focused on intrinsic sentence memorability. Here, we instead examine the effects of the particular context in which a given sentence appears within the experiment.

We found that the hit rate for a sentence was more consistent across participants (split-half

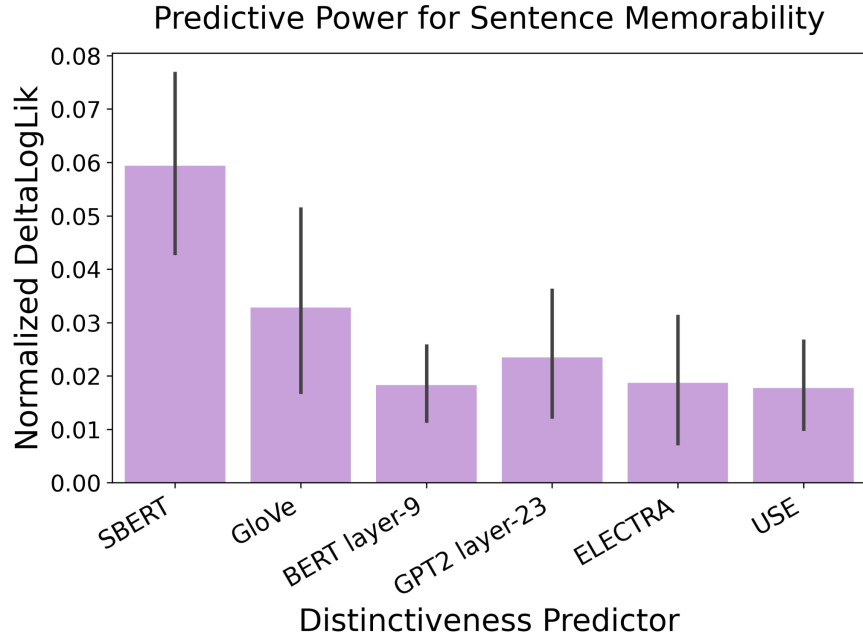


Figure 4: The effect of sentence-level semantic distinctiveness measures on sentence memorability. Delta Log-Likelihood (DLL) comparison for sentences in the DIVERSE condition ($n=1,000$). Values indicate the improvement in model fit when adding the given target predictor and an interaction term between the target predictor and word memorability, compared to a baseline model with only word memorability and word frequency. Error bars denote 95% CIs across 10 folds, each containing 10% of the data. Reported coefficients are based on a regression model fit on the full dataset.

Spearman’s $\rho = 0.56$) than the false alarm rate (split-half Spearman’s $\rho = 0.24$). One possible explanation for this difference is that the probability of a false alarm is more sensitive to the *specific* experimental sequence seen so far, which differs across participants. In particular, if memory representations are noisy, then encountering a sentence that is similar to a recently seen sentence (a distractor) may trigger a false memory.

As is commonly seen in recognition memory experiments, false alarms are initially infrequent, but rise with trial index, plateauing at around 5% (Figure 5A). Critically, we tested whether false alarms are affected by word-level overlap and sentence-meaning similarity with preceding sentences (see Section 2.5.5, Figure 5B and Figure 5C). Both critical predictors showed significant effects on correctness, with a higher probability of false alarms in the presence of lexical overlap and a semantically similar distractor (content word overlap: $\beta = -0.52$, $SE = 0.04$, $p < 0.001$; distractor cosine similarity: $\beta = -0.52$, $SE = 0.03$, $p < 0.001$). It is also worth noting that the distractor cosine similarity measure is not driven by lexical overlap: the closest semantic distractor only contained an overlapping content word with the current sentence 6.7% of the time.

In summary, in line with the noisy representation hypothesis, confusability with items in the recent context — both in terms of content word overlap and semantic similarity — increases the likelihood of errors due to a false impression that a sentence has been encountered before.

3.5 Intuitive judgments of sentence memorability align with empirical data

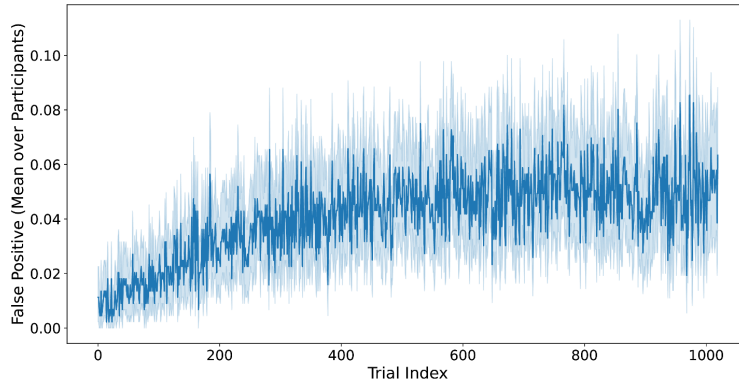
Finally, in an exploratory analysis, we tested whether participants have an intuitive sense of which sentences are memorable. In the domain of vision, participants have been shown to be poor at judging which images are memorable, and that these judgments have low consistency across participants (Bainbridge, 2017; Isola et al., 2014). In contrast, we find that participants **a)** are moderately consistent in their subjective ratings of memorability, as quantified using the same split-half approach as used for quantifying the consistency of sentence memorability scores (Spearman’s $\rho = 0.46$, 95% CI: [0.43, 0.50]), and **b)** can reliably, albeit far from perfectly, predict sentence memorability ($r = 0.24$, $p < 0.001$). These findings suggest that some degree of meaningful intuition exists about whether a sentence is likely to later be remembered.

We speculate that this difference from vision may be due to the fact that for images, subjective judgments of memorability are highly correlated with aesthetic judgments, while the actual memorability depends more on semantic image properties, such as the presence of a person, the location (e.g., indoors vs. outdoors), emotional valence, or actions taking place in the image (Isola et al., 2014). For sentences, the semantic content may be generally more salient because the very goal of linguistic messages is to convey a meaning. We speculate that extensive experience with communicative uses of language may also make participants more aware of how a given sentence may be processed and encoded by others, in contrast to vision, which frequently takes place outside the context of any communicative or social interaction.

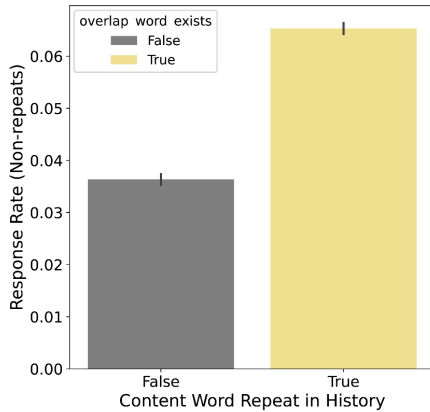
4 Discussion

In a large-scale sentence recognition memory experiment (2,500 target sentences, 443 participants), some sentences were recognized more accurately and faster than others, in a consistent

A False Alarm Rate by Trial Index



B False Alarm Responses Depending on Content Word Repeat in Sequence



C False Alarm Responses as a Function of Sentence Cosine Similarity in Sequence

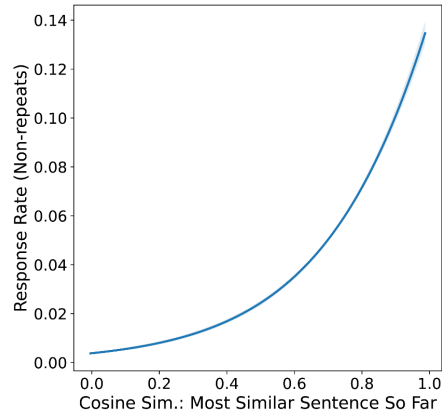


Figure 5: The effect of word-level and sentence-level properties on sentence confusability and false alarms. **(A)** False alarm rate (averaged across participants) by trial index. Error bars denote 95% confidence intervals. False alarm rate starts off close to 0, then trends upwards before reaching a plateau at around 0.05. **(B)** Responses on non-repeats (i.e., false alarms) depending on whether a content word in the current sentence has already appeared in the preceding experimental sequence. False alarms are more likely when a content word in the current sentence has previously appeared. **(C)** Logistic fit of responses on non-repeats (i.e., false alarms) as a function of SBERT cosine similarity between the current sentence and the most similar sentence in the preceding experimental sequence. As similarity to a preceding sentence increases, false alarms increase.

way across participants. In line with past work on visual memory (e.g., Brady et al., 2024) and word memorability (e.g., Tuckute et al., 2025), we found that sentences with more distinctive meanings — as estimated based on representations in large language models — were more memorable. Unlike previous works, which predicted memorability directly from high-dimensional semantic representations (Aka et al., 2023) or a set of semantic features (Madan, 2021), our distinctiveness metric is a scalar predictor computed from the *similarity* among high-dimensional representations; our metric thus operationalizes semantic distinctiveness as a single index of memorability, and we demonstrate that this unsupervised, theoretically motivated measure can predict empirical memorability. Different language models varied, however, in how useful their representations were for predicting memorability using the distinctiveness metric — SentenceBERT (Reimers & Gurevych, 2019), a model specifically trained to map sentences with similar meanings to nearby points in a semantic space, outperformed other models, such as ELECTRA (Clark et al., 2020) and USE (Cer et al., 2018). Other studies have found that embeddings based on SBERT outperform those based on ELECTRA and USE in semantic natural language processing tasks (Rep et al., 2024; Thakur et al., 2021), suggesting that SBERT embeddings more effectively capture sentence-level meanings. Moreover, although the presence of individually memorable words contributed to sentence memorability, the distinctiveness of sentence-level meanings explained variance beyond word-level memorability. Finally, aside from these intrinsic sentence memorability effects, we observed contextual effects: sentences that were preceded in the experiment by similar sentences were likely to trigger false alarms. Below, we elaborate on these findings and contextualize them with respect to broader issues in language and memory research.

4.1 The noisy representation hypothesis explains phenomena across domains and timescales

Previous studies of visual memory lend support to the noisy representation hypothesis — the idea that representations in human memory accumulate noise over time, making similar stimuli confusable (see Brady et al., 2024, for a review). This hypothesis offers an alternative to accounts that place a hard capacity limit on human visual working memory and construe memories in an all-or-none fashion (Awh et al., 2007; Cowan, 2001; Rouder et al., 2008; Todd & Marois, 2004; Zhang & Luck, 2008). Here, building on recent work on word memory (e.g., Tuckute et al., 2025; see also Dennis and Humphreys, 2001; Klaver et al., 2005; Steyvers and Malmberg, 2003; Walker and Hulme, 1999), we demonstrate that the noisy representation hypothesis can be extended to compositional linguistic stimuli: sentences. This account therefore appears broad in its scope, explaining effects across domains (visual and verbal stimuli) and stimulus complexity (single words and complex sentence-level stimuli). We note that this account is agnostic to whether the verbal and visual representation spaces are aligned; it is compatible with an aligned or shared semantic space (e.g., Huh et al., 2024; Roads & Love, 2020), but does not require it and would predict an effect of distinctiveness on memorability even in the case of distinct representation spaces. We note that, according to prior work, visual long-term memory can differentiate even small details, such as different views of the same object or slight changes in state (Brady et al., 2008); meanwhile, our results for sentences indicate a greater susceptibility to interference from similar sentences.

The noisy representation hypothesis also explains memory performance across timescales. In our experiment, memory performance was influenced by semantic distinctiveness arising from

both intrinsic properties of the sentences (i.e., how distinctive a sentence’s meaning is relative to a large corpus of sentences) and contextual factors (i.e., whether the experimental sequence contains sentences similar in meaning to the current target). According to the noisy representation hypothesis, there is no fundamental difference in representation between short-term and long-term memory (Brady et al., 2024; Schurgin et al., 2020), though there may be differences in the attentional resources needed to sustain the representations in short versus long-term memory. Indeed, the intrinsic memorability of a stimulus can be construed as its distinctiveness with regard to a much longer “context window”: in our case, the entirety of an individual’s prior linguistic experience. Rather than assuming that a participant begins the experiment with a “blank slate”, we might model a participant as coming in with a pre-existing backdrop of noisy sentence representations corresponding to the sum total of their linguistic experiences to date. Due to the statistical patterns present in language, speakers of the same language will have overlap in this longer-term set of encountered sentence meanings. Over the shorter temporal scale of the experimental session, a participant will encounter additional sentences. As a result, if a sentence falls in a dense part of the meaning space, a participant may be uncertain whether they saw this sentence earlier in the experiment, heard it on the television earlier that week, or read it in a book last year. In other words, the space of possible distractors includes both recently encountered experimental stimuli, but also prior linguistic experience more broadly. This construal can therefore explain both intrinsic memorability effects (due to generally shared linguistic experiences) and contextual memory effects (due to idiosyncratic experiences with particular stimulus sequences) in terms of the same memory processes.

4.2 Sentences are similar to words with respect to memory encoding

Past work on word memorability showed that words that have a unique association with a particular meaning (i.e., words with few synonyms and few meanings) are more memorable (Tuckute et al., 2025). Although in the current study, we operationalize semantic distinctiveness in a different way (using high-dimensional vector spaces), the results are conceptually parallel: words and sentences that are represented in sparser parts of the meaning space are more memorable. Similar cognitive processes may thus be at play whether the mind is representing the meaning of a discrete lexical item or the compositional meaning of a sentence.

This similarity of memory effects for words and sentences aligns with the general idea of a continuum of linguistic representations from morphemes and words to more complex syntactic constructions, long advocated by usage-based accounts of language (Bybee, 2010; Goldberg, 1995; Jackendoff, 2002; Tomasello, 2003). This idea has found empirical support in recent work in neuroscience and work on large language models. In particular, brain imaging investigations have shown that no sharp separation exists between brain areas that process word meanings and those that support combinatorial syntactic and semantic processing: instead, the same set of frontal and temporal brain areas show sensitivity to linguistic structure at all levels, from phonotactics, to word meanings, to syntax and compositional semantics (Bautista & Wilson, 2016; Caucheteux et al., 2021; Fedorenko et al., 2010, 2016, 2020; Regev et al., 2024; Shain et al., 2024). And large language models, which, as discussed in Section 1, have achieved impressive successes on diverse language tasks, use a single distributed representation space to encode both **a)** words/word-parts and **b)** syntactic patterns and complex meanings (Bölücü & Can, 2022; Ethayarajh, 2019; Hewitt & Manning, 2019). A continuum of linguistic representations in the human mind naturally predicts

similar memory encoding and retrieval effects for words and sentences.

4.3 Semantic composition: A sentence is more than the sum of its parts

Substantial variation in sentence memory performance (having to do with both intrinsic memorability and contextual effects) can be explained using word-level measures. However, the distinctiveness of sentence meanings explains additional variation, even when accounting for word-level features. Some sentences with comparable word-level memorability differ significantly in overall memorability depending on the distinctiveness of their compositional meanings. For example, the sentences “*We want to make it better.*” and “*One, two, three, four, five, six-*” have a similar average estimated word memorability, but the former sentence is one of the five least memorable sentences in the experiment, whereas the latter sentence is tied for being the most memorable. This example highlights how individually forgettable words can combine to create memorable meanings. Thus, although participants can and do use individual words to successfully recognize sentences, sentence memory performance cannot be reduced to word-level effects.

In the broader literature on semantic composition, it is well known that people can derive complex meanings from the composition of words even in novel combinations (Partee, 2004). The processing of sentences involves not only combining word meanings according to the sentence’s syntactic structure, but also draws on world knowledge and other sources of information (Gibson & Pearlmutter, 1998; Kintsch, 1988). The inferences that sentences afford can lead to more vivid and emotionally-charged meanings than those typically expressed by individual words (Zipf, 1949). For example, the sentence “*The boy went to sleep and never woke up again*” conveys a strongly negative meaning despite lacking negative-valence words (Lai et al., 2015). Similarly, we evidenced that compositional meanings of sentences influence their memory traces beyond their constituent words.

4.4 The importance of meaning does not imply that form is not important

So far, when discussing sentence distinctiveness, we have focused on the distinctiveness of sentence *meanings*. However, the same complex meaning can be expressed in a multitude of ways, and choosing the words and constructions so as to convey the intended meaning most precisely, compellingly, and in a way that evokes the right emotions and mental states is what great writers are known for. Does form-related distinctiveness affect our memory for sentences? The materials used in the current study may not be optimally suited to tackle this question because semantic and form-based distinctiveness are likely correlated in natural language. Moreover, the sentences were relatively short (all six words long), limiting the variability in syntactic structures. However, in an exploratory analysis (Section SI 8), we asked whether the unusualness of a sentence’s syntactic form may affect memorability. We used surprisal from a lexicalized probabilistic context free grammar (PCFG) parser, which is highly sensitive to unusual syntactic structures and uses of words in a different part of speech from their typical usage (van Schijndel et al., 2013). We compared PCFG surprisal against two other surprisal metrics: n-gram surprisal, which is sensitive only to local word sequences, and surprisal from the GPT-2 large language model (Radford et al., 2019), which captures many linguistic, form-based and semantic features (for reviews, see Chang & Bergen, 2024; Mahowald et al., 2024). Intriguingly, the structural PCFG surprisal fared best

among these three metrics at predicting sentence memorability, providing suggestive evidence that sentences with surprising syntactic constructions may be more memorable.

One interesting line of past work has investigated memory for form-based features in native vs. non-native speakers of a language and reported superior retention of lexical information in non-native speakers (Sampaio & Konopka, 2013). Future work may investigate sentence recognition memory in such populations testing whether these effects extend beyond lexical-level encoding, and whether the better retention of surface information may come at a cost of less efficient meaning encoding. Additionally, experiments with balanced bilingual participants could help test whether the representations of sentences in memory are agnostic to language, or whether the language-specific form in which a sentence is encountered is encoded in memory, even if the particular words and structures may be forgotten. If only the meaning is retained, then bilingual participants may be expected to experience distractor effects from sentences with similar meanings even when expressed in a different language.

4.5 Broader impact

The evidence in the current study attests to the explanatory power of the noisy representation hypothesis. Although originally proposed to explain visual memory, this hypothesis appears to offer a domain-general account of recognition memory based on the distinctiveness and confusability of noisy representations in memory. Furthermore, these behavioral findings open the door to neural investigations of how meanings are encoded and retrieved (Bainbridge & Rissman, 2018; Bogacz et al., 2001; Gershman et al., 2025; Mohsenzadeh et al., 2019; Wixted, 2007). Finally, and more generally, the ability to create memorable messages is important in diverse fields, from education, to politics, to science communication, to marketing, to arts and entertainment. These results can therefore inform efforts in these various spheres of life, to lead to more effective communication of information.

5 Author Contributions: CRediT Roles

Thomas Hikaru Clark: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Original Draft, Writing - Review and Editing, Visualization. **Greta Tuckute:** Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Original Draft, Writing - Review and Editing, Visualization. **Bryan Medina:** Conceptualization, Methodology, Software. **Evelina Fedorenko:** Conceptualization, Writing - Review and Editing, Funding Acquisition.

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SI 1 Exclusion criteria for experimental materials

When constructing experimental materials, sentences were omitted from consideration if (M=manual, A=automatic):

A: More than 50% of ‘words’ contain numeric characters.

A: More than 50% of characters are uppercase.

A: The sentence contains one or more ‘words’ that are longer than 20 characters.

A: More than 4 punctuation characters appear in a row.

M/A: The sentence contains more or fewer than 6 words due to word tokenization errors.

M/A: The sentence is cut off erroneously (e.g., due to a period after ‘Mr.’).

M: The sentence is Inappropriate/offensive (e.g., contains a racial slur or a taboo word).

M: The sentence contains several sub-sentences (i.e., two sentences in one marked by a period, an exclamation point, or a question mark)

M: Solely consists of a trademark/website name/brand or product name/person name or identifier/journal footer/solely date and place combination

M: Contains a type of emoji face, e.g., ‘:)’

M: Full utterance in foreign language

M: Erroneous sentences (e.g., agreement error)

M: Typos

During manual review, we standardized the materials: Unless a word was sentence-initial, a proper noun, or an acronym, it was lower-cased. Minor discrepancies in punctuation, such as sentences ending in commas or semicolons, were standardized. Obvious typos, misspellings, and highly nonstandard forms of words (such as *prob’ly* instead of *probably*) were also fixed manually.

SI 2 Classifying sentences based on word-level features

Step 1: Select a set of target words from existing data on word memorability.

Tuckute et al. (2025) report two large-scale experiments (Experiments 1 and 2) evaluating single word memorability, whose results suggest that words vary substantially in how memorable they are. Each of the two experiments contained approximately 2,200 critical words, but many ‘words’ consisted of multiple tokens (e.g., “LONG ISLAND ICED TEA” or “HIGH HEELS”). Here, we focused on a subset of 2,783 single-token words ($n=2,109$ items from Experiment 1 and $n=722$ from Experiment 2). Memorability scores were defined as response accuracy, which takes into account hits (correctly recognizing a repeat of a word) and false alarms (mistakenly selecting a word as a repeat on the first encounter, see Section 2.5.1). For the 2,783 words, we computed the mean and standard deviation of response accuracy. From this set, we then selected $n=1,428$ words that fell into three memorability conditions: high-memorability words ($n=461$) with accuracies at least 1 standard deviation above the mean, low-memorability words ($n=498$) with accuracies at least 1 standard deviation below the mean, and medium-memorability words ($n=469$) with accuracies within 0.25 standard deviations of the mean.

Step 2: Select a large set of sentences for consideration.

30,000 six-word-long sentences were sampled — 10,000 from each of the three corpus categories (Written-published, Written-web, Spoken) — with the criterion that each sentence contained at least one of the 1,428 “ground truth memorability” words (see Step 1). Some of the words in the HIGH memorability set were very common (“SHE”, “US”, and “AN”), present in $> 10\%$ of the sentences containing HIGH memorability words, and were therefore eliminated; similarly, one word in the MEDIUM memorability set was common (“DOWN”) and was also eliminated. An equal number of filler sentences (30,000) were sampled that did not contain any ground truth memorability words, resulting in a candidate stimulus pool of a total of 60,000 sentences.

Step 3: Train a linear regression model to predict memorability of any arbitrary word.

To estimate the memorability of all words present in the candidate target sentences, we used the ground truth word memorability dataset ($n=2,783$) to fit a linear regression model to predict a word’s memorability from a 300-dimensional GloVe embedding (GloVe: pre-trained vectors from the Common Crawl corpus, available at <http://nlp.stanford.edu/projects/glove/>, version glove.42B.300d.txt). The rationale for using GloVe embeddings is that words appear to be encoded into memory by their meanings rather than their surface forms (Begg & Wickelgren, 1974; Bransford & Franks, 1971; Franks & Bransford, 1972; Lombardi & Potter, 1992; Tuckute et al., 2025), and GloVe embeddings approximate word meanings well enough to predict judgments on diverse semantic tasks (Pereira et al., 2016).

The regression model can then take any word in the GloVe vocabulary (~1.9M words) and output an estimated word memorability score based on its semantic features as encoded in the GloVe embedding. Using 10-fold cross-validation, we found that a model trained on 90% of the word memorability data and evaluated on the remaining 10% achieved Pearson correlation coefficients with a mean value of 0.651 (median=0.648, SD = 0.039), indicating that the model captures a significant amount of the variance in true, experimentally estimated, word memorability. We therefore used a linear model trained on the full dataset for estimating the memorability of all words across the 60,000 sentences.

Step 4: Select the sentences for use in the experiment.

To estimate sentence-level memorability scores based on the memorability (actual or esti-

mated) of a sentence’s constituent words, we averaged the word memorability scores of all words in a sentence, ignoring those outside of the GloVe vocabulary (among the 60,000 sentences, most sentences contained no out-of-vocabulary words: median=0, mean=0.123). To select the target sentences from among the 30,000 sentences containing ground truth memorability words (600 for each of LOW, MEDIUM, and HIGH memorability conditions, with the goal of later sub-selecting 500 for each condition, as detailed below), we used the following criteria. A sentence was selected for the HIGH memorability condition if it **a)** contained at least one high memorability word (based on the empirical word memorability data of Tuckute et al., 2025), and **b)** had a mean estimated word memorability score that fell at or above one standard deviation above the mean estimated word memorability score computed over the full set of 60,000 sentences. A sentence was selected for the MEDIUM memorability condition if it **a)** contained at least one medium memorability word (and no high memorability words), and **b)** had a mean estimated word memorability score that fell within 0.25 standard deviations of the mean. Finally, a sentence was selected for the LOW memorability condition if it **a)** contained at least one low memorability word (and no medium or high memorability words), and **b)** had a mean estimated word memorability score that fell at or below 1 standard deviation below the mean.

Figure 6 summarizes the material selection procedure for the LOW, MEDIUM, and HIGH memorability conditions using a combination of ground truth memorability words and mean estimated word memorability scores.

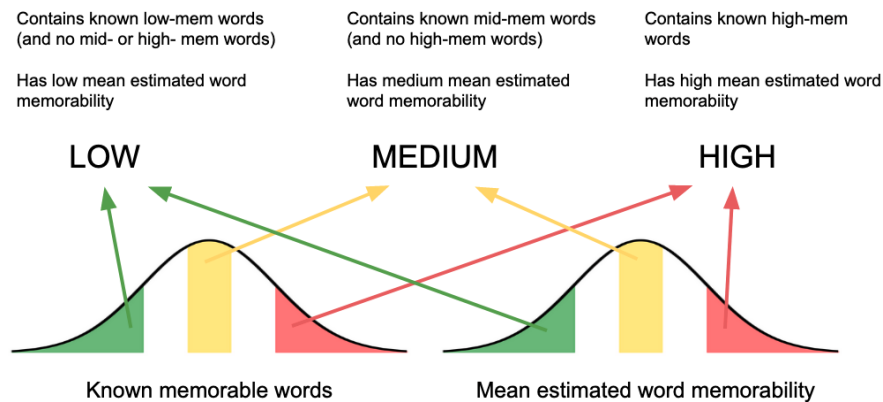


Figure 6: Summary of the criteria for sampling sentences for the three critical conditions: LOW, MEDIUM, and HIGH memorability.

We used the following values to define the categories for **the ground truth memorability** words: 1 standard deviation above or below the mean for HIGH and LOW, respectively, and within 0.25 standard deviations of the mean for MEDIUM. Words with an accuracy above 0.857 were considered HIGH memorability, while words with an accuracy below 0.737 were considered LOW memorability. Words with an accuracy between 0.782 and 0.812 were considered MEDIUM memorability.

We used the following values for defining the categories for **estimated word memorability** scores (averaged across words in a sentence): 1 standard deviation above or below the mean for HIGH and LOW, respectively, and within 0.25 standard deviations of the mean for MEDIUM.

Sentences with an estimated word memorability score above 0.780 were considered HIGH

1257 memorability. Sentences with an estimated word memorability below 0.742 were considered
1258 LOW memorability. Sentences with an estimated word memorability between 0.756 and 0.766
1259 were considered MEDIUM memorability.

SI 3 Specific hypotheses about the influence of word memorability on sentence memorability

We hypothesized that the memorability of a sentence depends, in part, on the memorability of its constituent words. We pre-registered the following sub-hypotheses regarding how word-level memorability may determine the memorability of a sentence, which we also illustrate schematically in Figure 7:

- A) **Mean memorability (equal weighting) hypothesis:** The memorability of a sentence depends on the memorability of all of its constituent words, with all words weighted equally. A higher average memorability score results in a more memorable sentence.
- B) **Max memorability hypothesis:** The memorability of a sentence depends on the most memorable word in the sentence. The presence of additional high-memorability words does not play a significant role in increasing the overall memorability of the sentence.
- C) **Memorable start hypothesis:** The memorability of a sentence depends on the memorability of the word(s) at the beginning of the sentence.
- D) **Memorable end hypothesis:** The memorability of a sentence depends on the memorability of the word(s) at the end of the sentence.
- E) **Threshold hypothesis:** Sentence memorability increases monotonically as the number/memorability of words that pass a certain threshold increases. Hence, the more words in the sentence pass the threshold (and/or the higher the average memorability is of the words passing the threshold), the more memorable the sentence is. The words that fall below the threshold do not play a role.

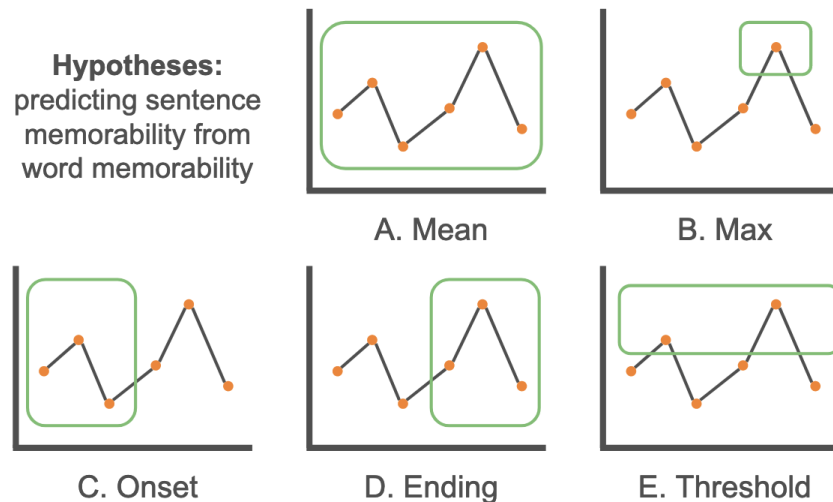


Figure 7: Schematic of the different ways in which the constituent words of a sentence could affect memorability of a sentence.

Although each of the predictors associated with each of the word-level memorability hypotheses show some support from the results, we note differences in how much each predictor

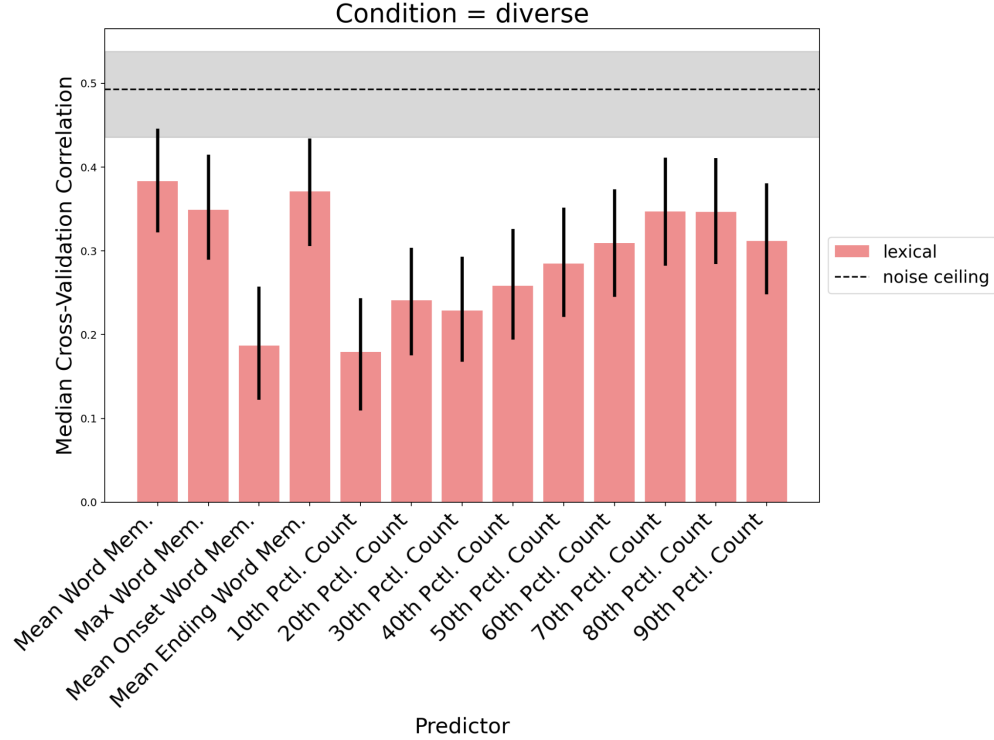


Figure 8: Systematic comparison of lexical predictors on sentence memorability. Cross-validated predictivity for sentence memorability, across a range of predictors based on estimated word memorability, on the Diverse sentence set ($n=1,000$ target sentences). Horizontal dashed line denotes the inter-participant reliability of the data, as defined by the split-half Spearman’s ρ for accuracy across participants, which measures the reliability of the data and hence provides a reasonable upper bound on the correlation obtainable from external predictors. Error bars indicate the 95% confidence interval estimated using 1,000 random splits of the data.

correlates with memorability. First, we observed that the mean memorability of the words in a sentence (Hypothesis 1) provided the numerically highest cross-validation scores, and hence chose to use this predictor as the lexical feature to compare to the sentence-level predictors in the main text (Section 3.3). Second, we noticed a large difference between the cross-validated scores for the memorability of the onset of a sentence (first three words in a sentence) and the cross-validated scores for the sentence ending (last three words in a sentence). We speculate that this difference is due to the higher rate of function word usage in sentence onsets compared to sentence endings. Third and finally, we found that the cross-validation scores generally increase when the predictor includes an increasing number of words that pass a certain memorability threshold, yet this effect appears to reach a plateau around the 70th percentile.

SI 4 Memorability performance broken down by condition

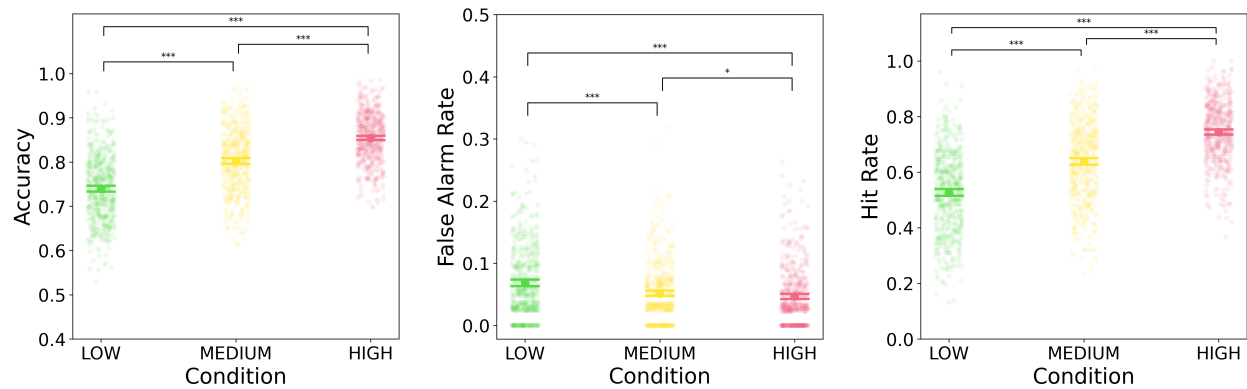


Figure 9: Accuracy, False Alarm Rate, and Hit Rate across conditions. Each translucent point denotes a sentence. Bold points denote condition means, and error bars denote 95% confidence intervals. The vertical axis of each plot measures a different performance metric. For the metric of false alarm rate only, smaller values indicate better performance. Regardless of metric, the LOW condition had the worst performance, the HIGH condition had the best performance, and MEDIUM lay in between. Pairwise differences were all significant by the Mann-Whitney U test, with $p < 0.001$ in all comparisons except for between MEDIUM and HIGH in False Alarm Rate, where $p < 0.05$.

SI 5 Most and least memorable sentences by condition

Condition	Most Memorable	Least Memorable
HIGH	Wait a sec, Nicky, Nicky, Nicky. Homer Simpson is hungry, very hungry. Does olive oil work for tanning? These mosquitos are – well, guinea pigs. Goddammit, where are you from, Mars? Macbook Pro repair in Niagara region. Now let's talk about Donkey Kong. Top with mozzarella, mushrooms, and peppers. The guy spelled sex and danger. San Diego, for Jay Leno - hello.	Tiny soup champagne volunteer personalized denver. You just go like this, dear. Where did that other guy go? Her unborn son died as well. I mean look at this sweetie. I'm doing this for my son. This Colleen Sutton woman I'm with. Jesse Fisher is a master craftsman. Seth hefted a coarse brown sack. But don't think it's over, Abby.
MEDIUM	They stabbed her fifty three times! Every cloud has a blue lining! The bigger the hair, the better. I trained people with plastic explosives. Let's individually start with Bruce Jenner. His gray hair was cut short. Everyone was looking for Michael Jordan. What is another word for poor? It was better than sex, Mark. Your parents came to my funeral.	And look what we have left. He says it's missing something important. Here is the third day out. And I had the afternoons free. And weak force during acquisitive accuracy. About the mark and my problems. It is a really nice idea. They date to the same year. I've no idea what will happen. Because this could mark their comeback.
LOW	For instance, to, too, and two. But isn't quick weight loss dangerous? Don't be afraid to be lazy. Actually, I already work for you. Create amazing logos quick and easily. Puppies are totally capable of learning. Can a bru53x replace a bru5g? He has a very powerful ship. Only serious managers need to apply. The key to survival is moderation.	I just couldn't leave without it. We want to make it better. But still it wasn't nearly enough. It must be getting more serious. But it made me feel better. Therefore to hold meetings by them. Another manner is to time entries. But give it to me again. You still had to prove yourself. Unless he'd been afraid they'd know.
DIVERSE	Do you have 2 memory DIMMS? One, two, three, four, five, six- Sizes range from 2XL to 3XL. A cigarette would taste particularly good. Take 1 leaf of Aloe Vera. What Does 'Surf the Web' Mean? I highly recommend the hot chik'n. Light is made up of photons. Amazing place to learn martial arts. Press F5 to reset the position.	I thought that you were anyway. You bet there's other things here. We may outgrow and forget them. I don't have the answer yet. I have to ask you that. And that is a direct strategy. But then again he could be! That could be a commendable quality. I feel the same way sometimes. You're the one who started it.

Table 1: Most and least memorable sentences by condition. For each condition, the 10 sentences with highest response accuracy (left column) and the 10 sentences with lowest response accuracy (right column) are shown.

SI 6 Full regression model results

Tables 2 to 7 show regression model results for predicting accuracy from semantic distinctiveness (linear regression). Table 8 shows regression model results for predicting response times from distinctiveness at the trial level (linear mixed-effects regression). Table 9 shows regression model results for predicting correct responses (a binary outcome) using contextual and intrinsic predictors (logistic mixed-effects regression). We use the following abbreviations: word_freq denotes word frequency (from SUBTLEXus); word_mem denotes mean estimated word-level memorability; distinct_{MODEL} denotes distinctiveness in the embedding space of a given model, e.g. GloVe or SBERT; sentence_nchar denotes the number of characters in a sentence; repeat is a binary variable denoting whether a sentence has previously appeared (1) or not (0); distractor_cosine_sim denotes the cosine similarity in the SBERT space between the current sentence and the most similar sentence in the experimental history; content_word_overlap is a binary variable denoting whether a content word from the current sentence has previously appeared in the experiment (1) or not (0).

	Model 1
(Intercept)	0.0880 (0.0296)**
word_freq	−0.0615 (0.0499)
word_mem	0.4866 (0.0410)***
distinct_glove	0.2199 (0.0593)***
word_mem:distinct_glove	−0.1864 (0.0325)***
R ²	0.2328
Adj. R ²	0.2297
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 2: distinct-glove as predictor

	Model 1
(Intercept)	0.0498 (0.0295)
word_freq	−0.1263 (0.0307)***
word_mem	0.3437 (0.0399)***
distinct_sbert	0.2834 (0.0300)***
word_mem:distinct_sbert	−0.1611 (0.0336)***
R ²	0.2819
Adj. R ²	0.2791
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 3: distinct-sbert as predictor

	Model 1
(Intercept)	0.0632 (0.0284)*
word_freq	−0.1796 (0.0325)***
word_mem	0.4663 (0.0383)***
distinct_gpt2	0.0585 (0.0312)
word_mem:distinct_gpt2	−0.1633 (0.0343)***
R ²	0.2229
Adj. R ²	0.2198
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 4: distinct-gpt2 as predictor

	Model 1
(Intercept)	0.0455 (0.0283)
word_freq	−0.1332 (0.0321)***
word_mem	0.4502 (0.0381)***
distinct_bert	0.1703 (0.0326)***
word_mem:distinct_bert	−0.0699 (0.0398)
R ²	0.2266
Adj. R ²	0.2235
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 5: distinct-bert as predictor

	Model 1
(Intercept)	0.0477 (0.0283)
word_freq	−0.2425 (0.0371)***
word_mem	0.4920 (0.0388)***
distinct_use	−0.0982 (0.0342)**
word_mem:distinct_use	−0.0995 (0.0370)**
R ²	0.2140
Adj. R ²	0.2108
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 6: distinct-use as predictor

	Model 1
(Intercept)	0.0450 (0.0281)
word_freq	−0.1895 (0.0324)***
word_mem	0.5034 (0.0387)***
distinct_electra	0.0019 (0.0285)
word_mem:distinct_electra	−0.1202 (0.0294)***
R ²	0.2170
Adj. R ²	0.2138
Num. obs.	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 7: distinct-electra as predictor

	Model 1
(Intercept)	1315.0853 (23.0816)***
word_mem	−45.9784 (3.1347)***
word_freq	−22.2976 (3.8788)***
sentence_nchar	9.8189 (0.5776)***
‘repeat’	−199.1606 (5.6929)***
distinct_sbert	−36.0084 (2.8930)***
AIC	795196.9423
BIC	795277.0451
Log Likelihood	−397589.4712
Num. obs.	54193
Num. groups: sentence	2500
Num. groups: worker_id	443
Var: sentence (Intercept)	8364.5935
Var: worker_id (Intercept)	51810.5561
Var: Residual	128631.5247

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 8: Response Times

	Model 1
(Intercept)	3.3661 (0.0564)***
word_freq	−0.0103 (0.0237)
word_mem	0.0092 (0.0242)
distractor_cosine_sim	−0.5201 (0.0315)***
content_word_overlap	−0.5238 (0.0380)***
AIC	34356.3912
BIC	34422.0123
Log Likelihood	−17171.1956
Num. obs.	87067
Num. groups: sentence	2500
Num. groups: worker_id	443
Var: sentence (Intercept)	0.4035
Var: worker_id (Intercept)	0.6740

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 9: False Positives - Logistic Regression

SI 7 Comparison of BERT and GPT-2 layers

For GPT-2 and BERT, we extracted model activations from each of the models' 25 layers (24 hidden layers and 1 input embedding layer). We compare the predictivity of a range of choices for the model layers for GPT-2 and BERT. We evaluate on the DIVERSE condition, in order to span a diverse range of naturalistic stimuli without specifically controlling word-level memorability. We find that later layers in GPT-2 and middle layers in BERT generally perform better at predicting memorability when used as a basis for the distinctiveness measure.

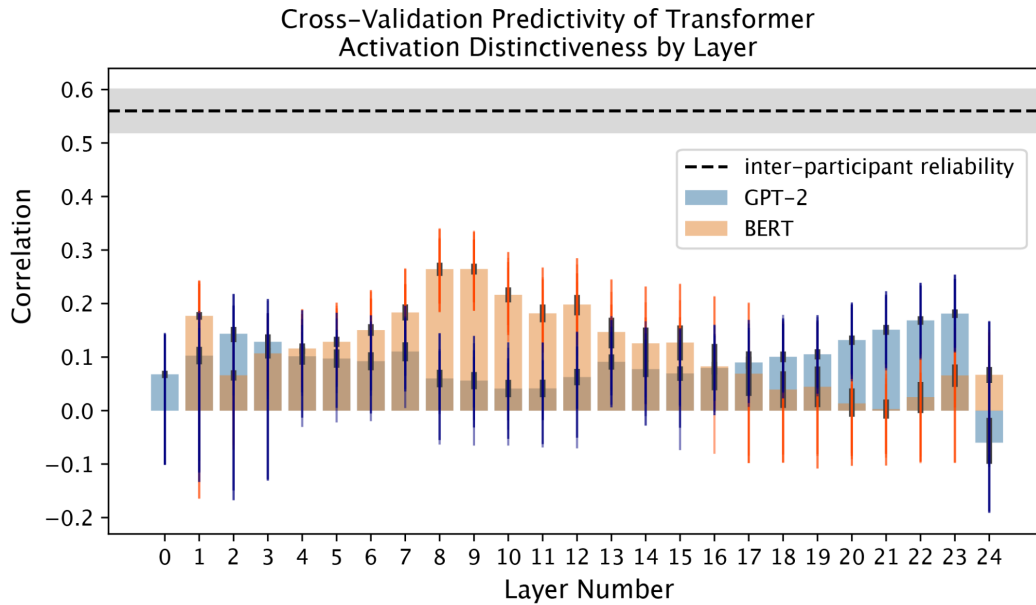


Figure 10: Cross-validated predictivity of Transformer activation distinctiveness by layer in the DIVERSE condition. Per-layer predictivity scores for response accuracy. Higher values for a layer number indicate that the distinctiveness metric for a sentence computed using the model activations from the specified layer of either the GPT-2 or BERT model is a better predictor of sentence memorability. The dashed line indicates the inter-participant reliability of response accuracy. GPT-2 and BERT exhibit different patterns, with BERT showing the best predictivity in intermediate layers and GPT-2 showing best predictivity in the penultimate layer.

SI 8 Effect of three surprisal predictors on memorability

We conducted an additional analysis investigating the relationship between three types of contextual surprisal and sentence memorability. Surprisal is an information-theoretic measure of how difficult it is to predict each word in a sentence, given the preceding context within that sentence. Surprisal has also been shown to be a robust predictor of language processing effort, as measured by reading times (e.g., Brothers & Kuperberg, 2021; Smith & Levy, 2013) or level of activity in the brain’s language network (e.g., Shain et al., 2020; Willems et al., 2016).

We included three different surprisal models: n-gram surprisal is indicative of local surprisal based on particular word sequences, lexicalized PCFG surprisal is sensitive to part of speech patterns (van Schijndel et al., 2013), and GPT-2 surprisal likely includes all features that are relevant to determine which word is likely to come next (Chang & Bergen, 2024; Mahowald et al., 2024; Radford et al., 2019). We describe each surprisal model below.

N-gram surprisal (5-gram): We computed smoothed n-gram surprisal using a 5th order n-gram language model, which estimates surprisal given a context window of up to four preceding words. The model was trained using the KenLM library (Heafield, 2011) and built on 12/24/2022 on the training set of Wikitext-103 (Merity et al., 2016), with default smoothing parameters (modified Kneser-Ney smoothing) to handle sparsity. The training data were tokenized using the sent_tokenizer from the NLTK library (Bird et al., 2009; version 3.8) followed by tokenization of sentences using the wordpunct_tokenizer from the same library. Punctuation was stripped and all words were lower-cased. The same tokenization and preprocessing procedure was applied to the sentence materials. We obtained a surprisal estimate for each word in each sentence as well as an end-of-sentence token (</s>) using default parameters of the full_scores() function in KenLM, and then aggregated surprisal additively resulting in a single estimate per sentence. The following sentences had the highest n-gram surprisal across all experimental materials: *“Tubs, cartons, soap pumps, shampoo bottles.”*, *“Stamping tink: September pencil class - flowers!”*, *“Jasmine coloring pages princess free printable.”*, *“HTML convert time to 0.264 sec.”*, *“Dear Governor Bevin, I’m Audra Baker.”*

Probabilistic context-free grammar (PCFG) surprisal: We computed lexicalized probabilistic context-free grammar (PCFG) surprisal using the incremental left-corner parser of van Schijndel et al. (van Schijndel et al., 2013) trained on a generalized categorial grammar (Nguyen et al., 2012) reannotation of Wall Street Journal sections 2 through 21 of the Penn Treebank (Marcus et al., 1993). Each sentence was tokenized using a Penn Treebank Tokenizer and punctuation was retained. We obtained a surprisal estimate for each word in each sentence and then aggregated surprisal additively resulting in a single estimate per sentence. The following sentences had the highest PCFG surprisal across all experimental materials: *“One, two, three, four, five, six-”*, *“Reply: Dale Worley: RE: external error.”*, *“NOTD: Collection 2000 hot looks ‘Diva’.”*, *“Valuation question: Harper Rookie ‘Hit Balls’.”*, *“My pineapple coconut martini was delicious!!!”*

GPT-2 surprisal: We computed surprisal values using the pretrained, autoregressive GPT-2 language model available via the HuggingFace library (Wolf et al., 2020). GPT-2 surprisal likely includes all features that are relevant to determine which word is likely to come next Chang and Bergen (2024), Mahowald et al. (2024), and Radford et al. (2019). We obtained a surprisal estimate for each token in each sentence and then aggregated surprisal additively resulting in a single estimate per sentence. The following sentences had the highest GPT-2 surprisal across all experimental materials: *“Alexander Eidenier, Alvordton, Ohio, speed, 130.”*, *“Gary Strieker reports from*

Kotzebue, Alaska.”, “Tantalum, Titanium, Hastelloy, Inconel and Monel.”, “In China Avalokiteshvara becomes Kuan Yin.”, “Robert Boochever, Juneau, Alaska, for appellees.”

We compare the predictive power of each surprisal measure using the Delta log-likelihood analysis. We test the increase in model log-likelihood due to adding a given predictor to a baseline model predicting memorability from only average word frequency and average estimated word memorability. PCFG surprisal emerged as the best among the surprisal predictors at predicting sentence memorability. A possible interpretation of this result is that sentences with surprising syntactic constructions are more memorable. The sentence with the highest PCFG surprisal was “One, two, three, four, five, six-”, which has unusual syntactic structure (just a sequence of numerals), despite being unsurprising in other ways; this sentence was also tied for being the most memorable sentence in the experiment.

Sentence surprisal, computed using the above models, was positively correlated with SBERT distinctiveness (SBERT and GPT-2: 0.19; SBERT and PCFG: 0.23; SBERT and n-gram: 0.18). In theory, surprisal and distinctiveness can be dissociated: the sentence “My son was outside playing soccer.” ranked very highly on distinctiveness (4th-highest SBERT distinctiveness value out of 2,500 sentence stimuli) but relatively low on surprisal (1664th-highest GPT-2 surprisal value out of 2,500 sentence stimuli). In this study, naturalistic sentences from existing corpora were chosen to capture a broad range of sentence features. Future work can consider targeted manipulations that disentangle various parts of surprisal and semantic distinctiveness.

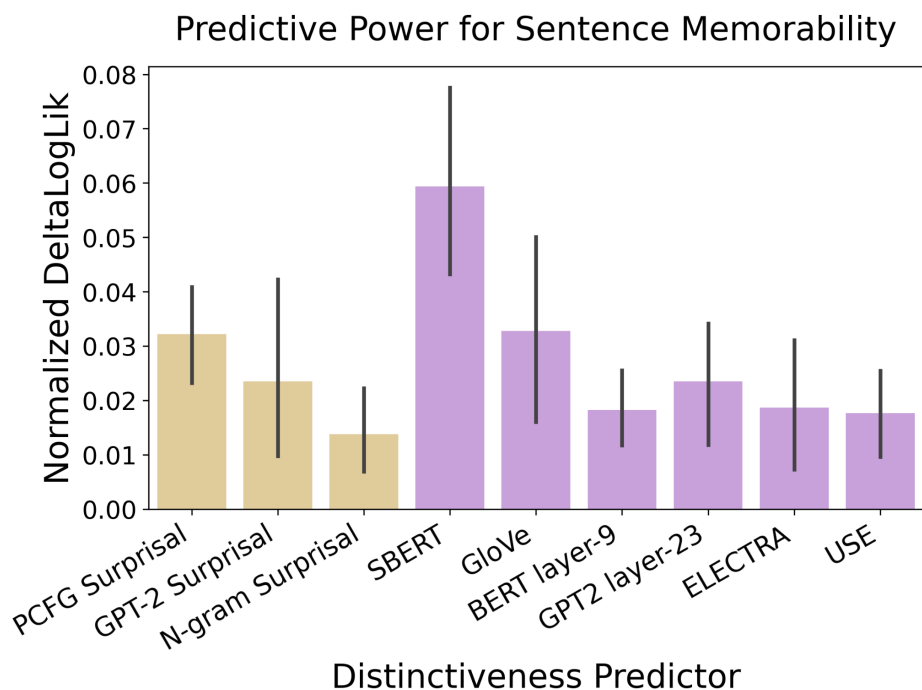


Figure 11: Delta Log-Likelihood for different semantic distinctiveness and surprisal predictors, compared to a baseline model including only average word frequency and average estimated word memorability. Higher values indicate better predictive power. Error bars denote 95% CIs across 10 folds of data. DLL was computed based on models fit with linear regression, predicting mean accuracy across participants.

SI 9 Intuitive judgments of memorability

Experimental instructions and participants

The subjective memorability norms for the DIVERSE sentences ($n=1,000$) were collected as part of a larger experiment consisting of a total of $n=2,000$ sentences (Tuckute et al., 2024). Each participant was exposed to a random subset of 100 sentences.

Instructions In this survey, you will be asked to rate 100 sentences. We would like you to rate each sentence according to **how memorable you think the sentence is** on a scale from 1 (not at all) to 7 (very much). If you **think that you would remember that you have encountered the sentence if asked again in a few hours**, you should rate the sentence as 6 or 7. In contrast, if you think that you would forget that you have encountered the sentence, you should rate the sentence as 1 or 2.

The question phrasing for each item was: “How memorable do you think the sentence is?”

Participants 400 participants took part in the experiment, recruited via the Prolific crowdsourcing platform. 107 participants were excluded following pre-defined exclusion criteria (see below), leaving 293 participants (73.25%). In particular, 19 participants were excluded because they listed a language other than English as their native language, 59 participants were excluded based on sentence completions, 6 participants were excluded based on their average response times, 1 participant was excluded based on the lack of rating variance, and 22 participants were excluded based on a low correlation with the remaining participants in their experimental list. The experiment took 13.76 minutes, on average ($SD = 2.03$). Each item was rated by 12-17 participants (14.65 participants on average, $SD = 1.60$). We note that the number of participants is lower than in the critical sentence memorability experiment, but financial constraints limited the recruitment to a maximum of 400 participants in the norming studies. The experiments were conducted with approval from and in accordance with the Committee on the Use of Humans as Experimental Subjects at the Massachusetts Institute of Technology. Participants gave informed consent before starting each experiment and were compensated for their time (minimum US\$12/hour).

Participant exclusion criteria

1. **Native speaker status:** Participants were excluded based on their native speaker status self report as well as the Prolific language and location filters.
2. **Sentence completions:** Participants were excluded if their sentence completions were ungrammatical, contained spelling errors (that were not obvious typos) or if the completions were deeply nonsensical. For the sentence completions, participants were instructed: “Please finish the following sentences:”
 - i) The garden that ...
 - ii) When I was younger, I would often ...
 - iii) The longer the workers protested, ...

iv) I could never have imagined that ...

v) Because Jane lived by herself, ...

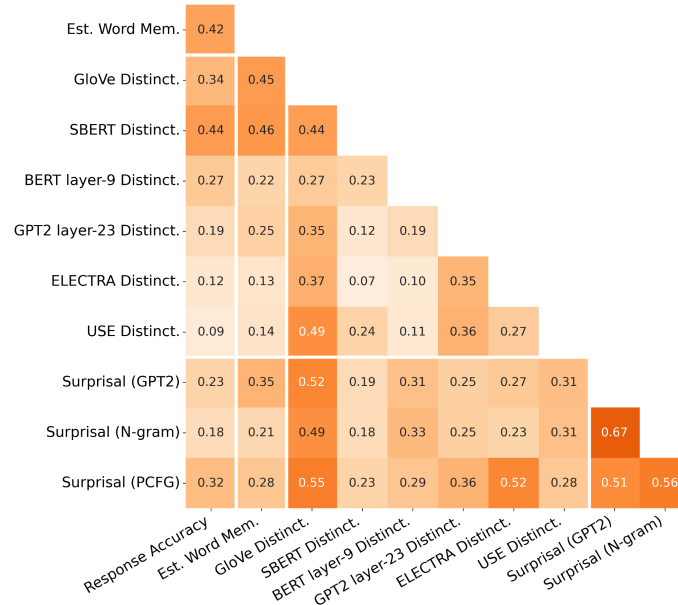
vi) The most difficult thing about the trip was that ...

The completions were judged for grammaticality by two different raters.

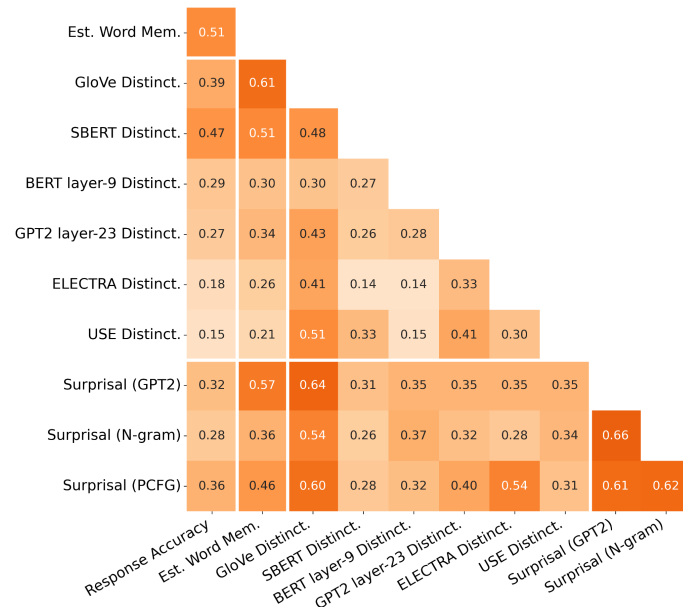
3. **Average response time:** Participants were excluded if the average response time per question was less than 3 seconds (i.e., for the survey that contained two questions, the threshold was 6 seconds).
4. **Lack of variance in ratings:** Participants were excluded if they only used a total of 2 unique rating values (out of 7) for all items in the survey.
5. **Correlation with other participants:** Participants were excluded if the average Pearson correlation with the ratings of remaining participants fell below 2 standard deviations below the mean inter-participant correlation. The inter-participant correlation was computed by correlating a vector of responses for a given participant with the vector of responses for each of the remaining participants and taking the average of these pairwise correlation values.

SI 10 Predictor correlations

The heatmaps in Section SI 10 show the correlation between sentence features. We observe that the two features with the highest raw correlation with mean response accuracy are estimated word memorability and SBERT distinctiveness.



(a) Correlation between predictors, and correlation with response accuracy, across 1,000 DIVERSE sentences.



(b) Correlation between predictors, and correlation with response accuracy, across all 2,500 target sentences (comprising 1,000 DIVERSE sentences and 500 sentences in each of the HIGH, MEDIUM, and LOW conditions).